

Analysis group meeting

Mathieu Ouillon (Mississippi State University)

09/15/25

- Electron identification:
 - The identification process occurs in two stages: (1) primary identification within the Event Builder reconstruction framework, and (2) analysis-level refinements to optimize sample purity for physics measurements.
 - The primary challenge in electron identification involves discriminating against minimum ionizing particles (MIPs), predominantly negative pions (π^-), which can mimic electron signatures in certain detector subsystems. This discrimination is achieved through the use of Cherenkov radiation detection and electromagnetic calorimetry.
- This is based on common analysis notes: [RG-A](#), [RG-M](#), and [RG-K](#).
- This analysis utilizes Pass1 data for outbending configuration and Pass0v11 data for inbending configuration

RG-D: Electron identification

- The Event Builder applies a set of criteria to negative tracks to assign electron particle identification (PID = 11). The selection requirements are:

Cut	Requirement	Purpose
Charge	$q = -1$	Select negative tracks
HTCC Photoelectrons	$N_{\text{phe}} > 2$	Reject charged hadrons
PCAL Energy	$E_{\text{PCal}} > 60 \text{ MeV}$	Remove MIPs
Sampling Fraction	$\pm 5\sigma$ vs $E_{\text{Tot,dep}}$	Electromagnetic shower identification
Track Matching	Geometric matching	Ensure correct track

These cuts are already applied by the Event Builder in COATJAVA and are not modified for RG-D.

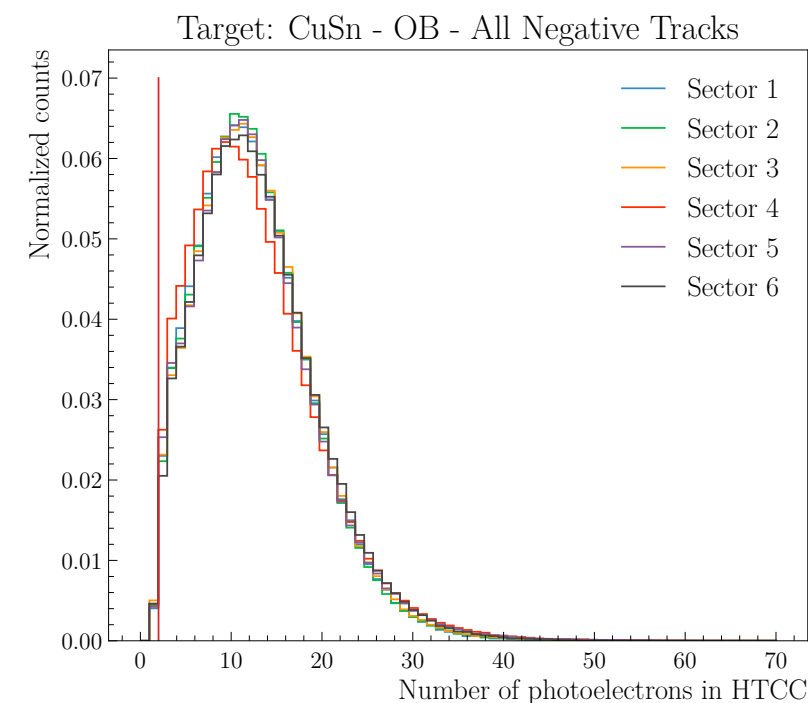
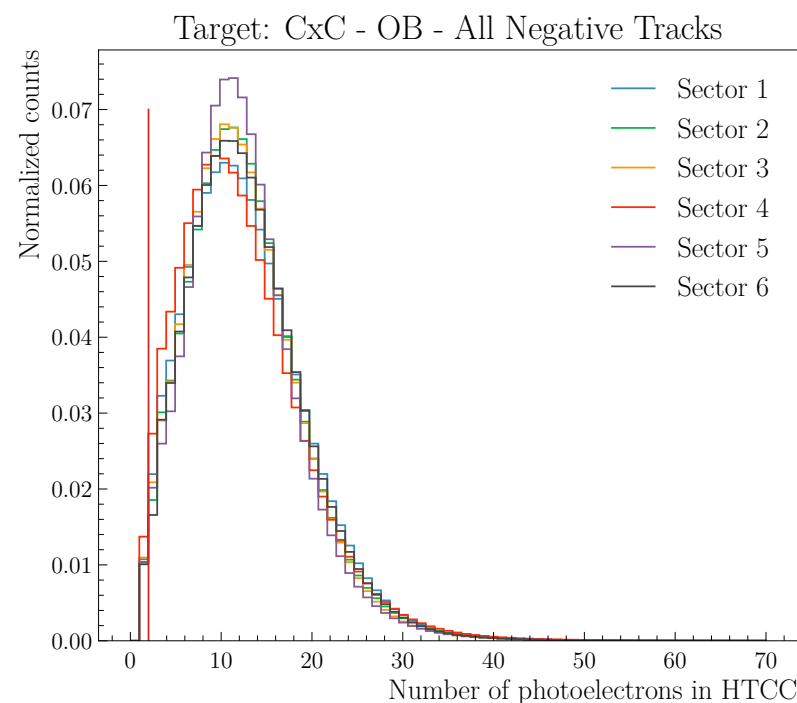
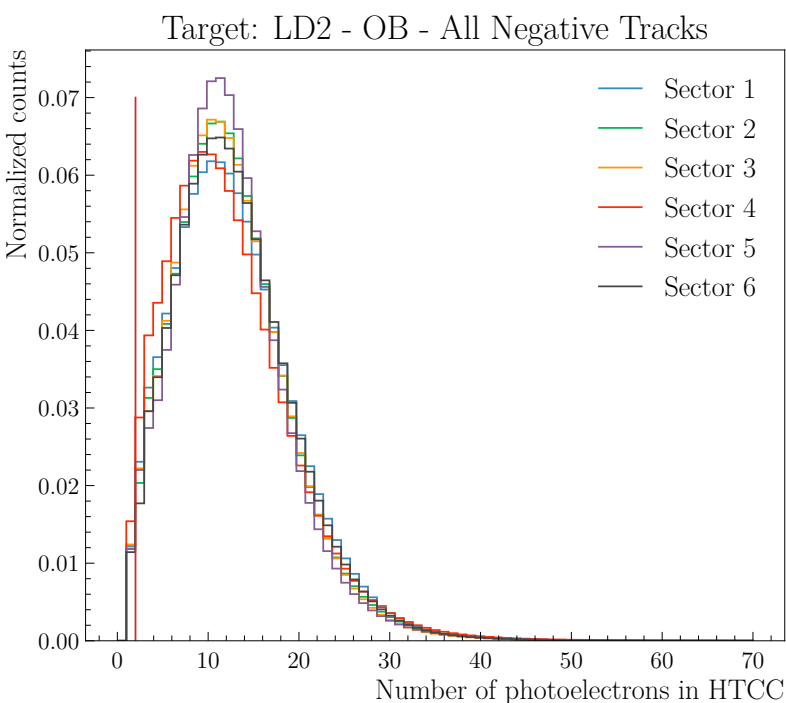
Electron Identification Refinements:

- RG-A: PCAL Minimum Energy Deposition Cut $E_{\text{PCal}} > 0.07 \text{ GeV}$ – Not used in RG-M, and RG-K (use default in the Event builder 0.06 GeV)
- RG-M: Sampling Fraction Cut vs momentum $\mu \pm 3\sigma$ – RG-A, and RG-K use this cut
- RG-M: Sampling Fraction Cut vs Total Energy deposited $\mu \pm 3\sigma$ – Not used in RG-A, and RG-K (use default Sampling Fraction vs Edep $\mu \pm 5\sigma$)
- RG-A/RG-K: Cut on the sampling fraction of the inner calorimeter and the PCAL - Not used in RG-M
- RG-M/RG-K: Cut on the electron momentum – Not defined in the RG-A common analysis note
- RG-A/K/M: vertex cuts

- Detector Status for trigger electron:
 - Electron candidates are restricted to the Forward Detector (FD) system through the detector status requirement:
 - $-4000 < \text{status} \leq -2000$
 - where negative values indicate trigger-electron assignment.
 - The status parameter encodes detector topology according to:
 - $\text{status} = 1000 \times \text{FT} + 2000 \times \text{FD} + 4000 \times \text{CD} + 100 \times \text{N_scint} + 10 \times \text{N_cal} + 1 \times \text{N_cher}$

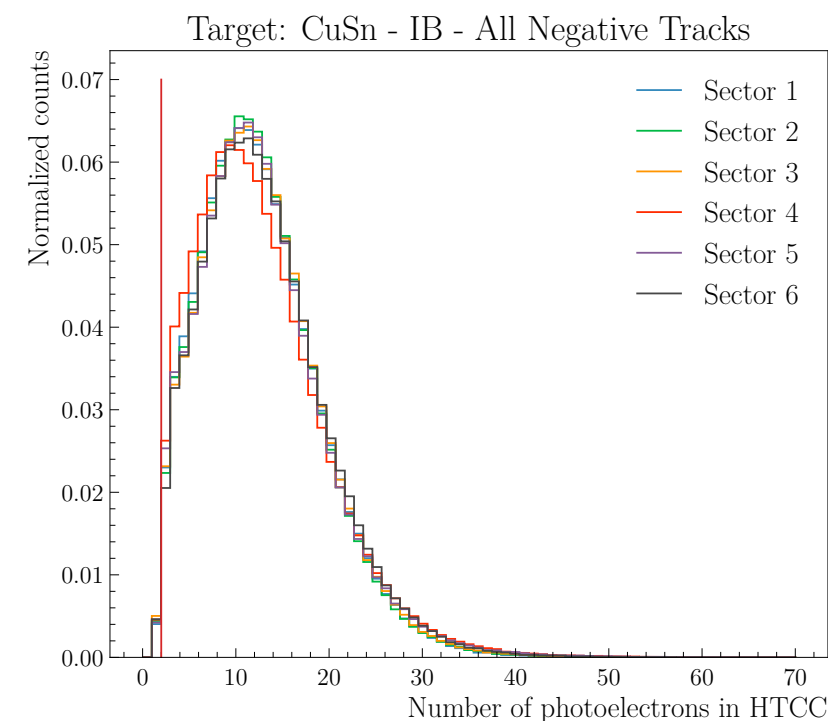
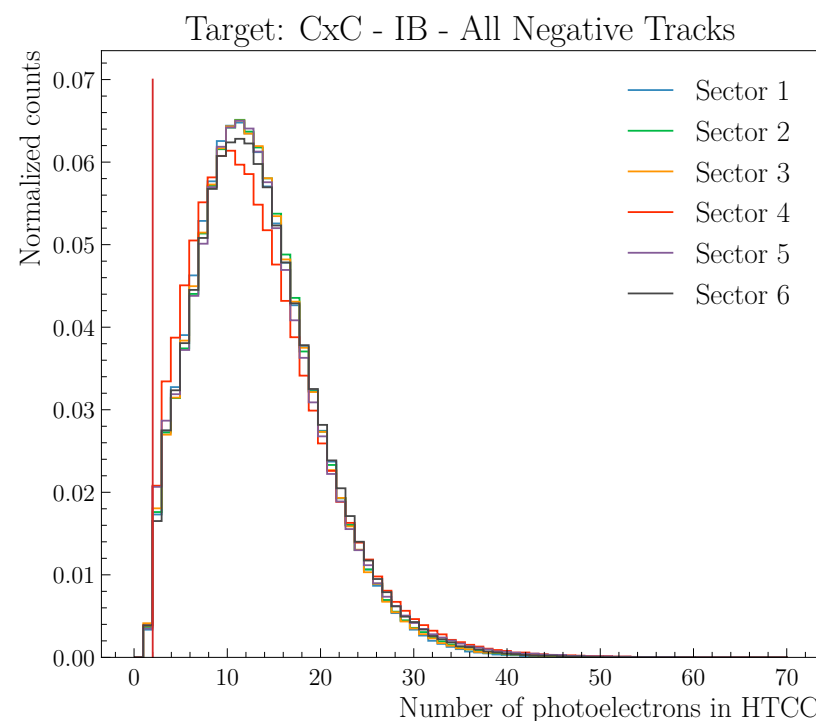
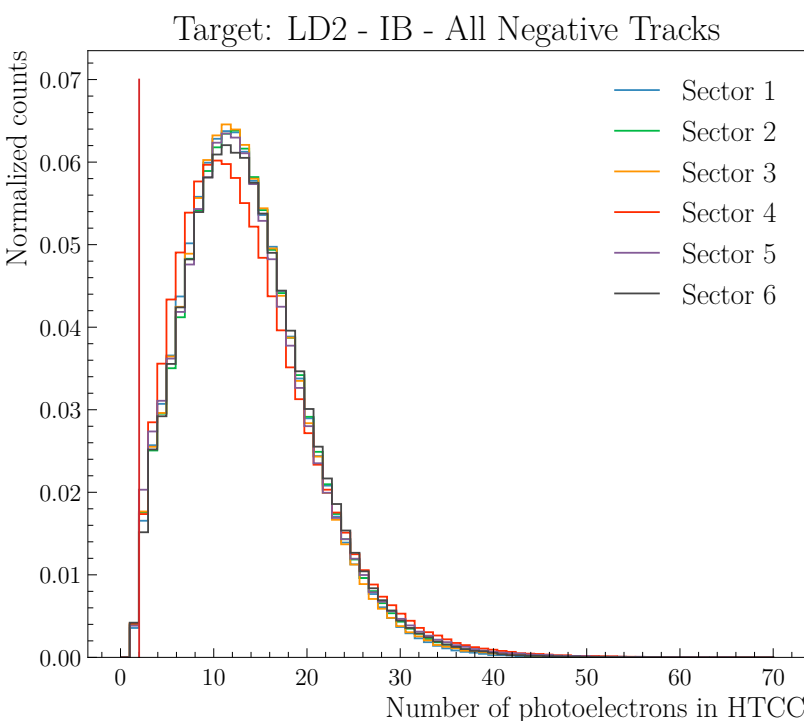
RG-D: Electron identification

- Electron Particle ID with Event Builder: Number of photoelectrons in the HTCC
 - **Purpose:** Minimize negative pion contamination in electron samples
 - **For electron candidate tracks:** below the pion threshold of 4.5 GeV, the HTCC effectively discriminates between electrons and pions by measuring the number of photoelectrons. Electron candidates produce more than 2 photoelectrons, which serves as the minimum threshold for particle identification cuts. Beyond that threshold pions begin producing Cherenkov light above threshold.
 - **Discrete hit distribution** - no fiducial cuts needed



RG-D: Electron identification

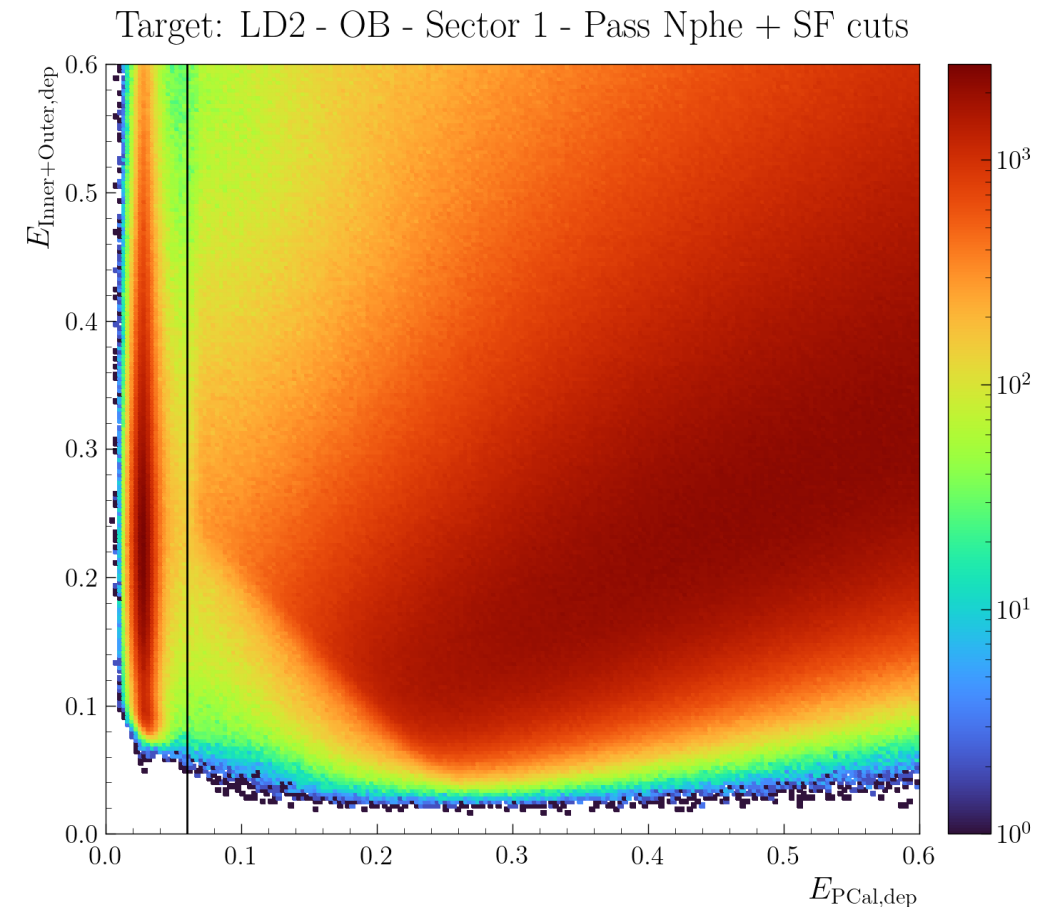
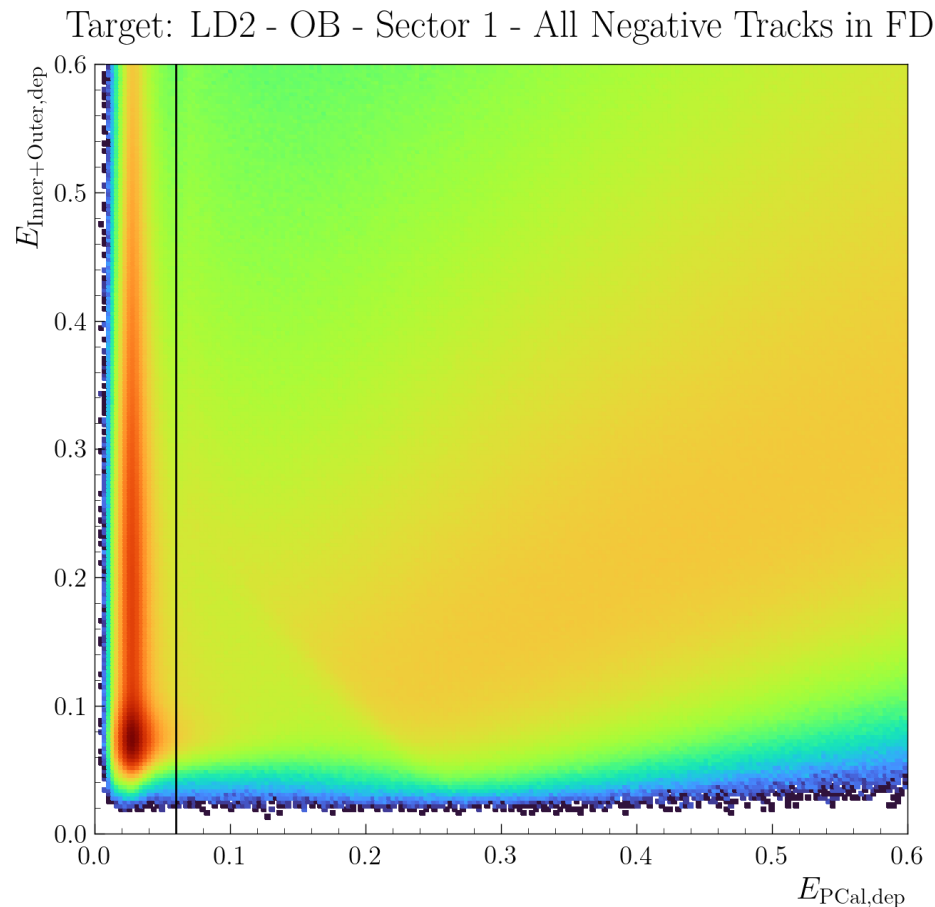
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RG-D: Electron identification

- Electron Particle ID with Event Builder: Minimum PCAL Energy Threshold Cut
 - The minimum energy deposition requirement $E_{\text{PCal}} > 60 \text{ MeV}$ exploits the difference between the electromagnetic shower of electrons and the minimum ionizing particles (pions). Electrons develop extended electromagnetic showers with significant energy deposition, while pions deposit a much smaller amount.

Black line is
0.06 GeV cut



RG-D: Electron identification

- Electron Particle ID with Event Builder: Sampling Fraction vs E_{dep} Cut

- The sampling fraction is defined as the ratio of the total deposited energy to the momentum

$$\text{SF} = \frac{E_{\text{PCAL,dep}} + E_{\text{Inner,dep}} + E_{\text{Outer,dep}}}{p}$$

- For electrons, this fraction remains nearly constant at ~ 0.25 across all momenta, indicating that the deposited energy scales proportionally with the electron's momentum. In contrast, negative pions, as MIPs, deposit a relatively constant amount of energy regardless of momentum.
- To accurately select electrons, tracks are required to be within $\mu \pm 5\sigma$ of the sampling fraction as a function of total deposited energy.
- This is achieved by binning the data in $E_{\text{Tot,dep}}$, fitting a Gaussian to the sampling fraction distribution in each bin to extract the mean (μ_{SF}) and standard deviation (σ_{SF}), and then parameterizing these values using:

$$\mu_{\text{SF}} = a(b + c/E_{\text{Tot,dep}} + d/E_{\text{Tot,dep}}^2)$$

$$\sigma_{\text{SF}} = e(b + f/E_{\text{Tot,dep}} + g/E_{\text{Tot,dep}}^2)$$

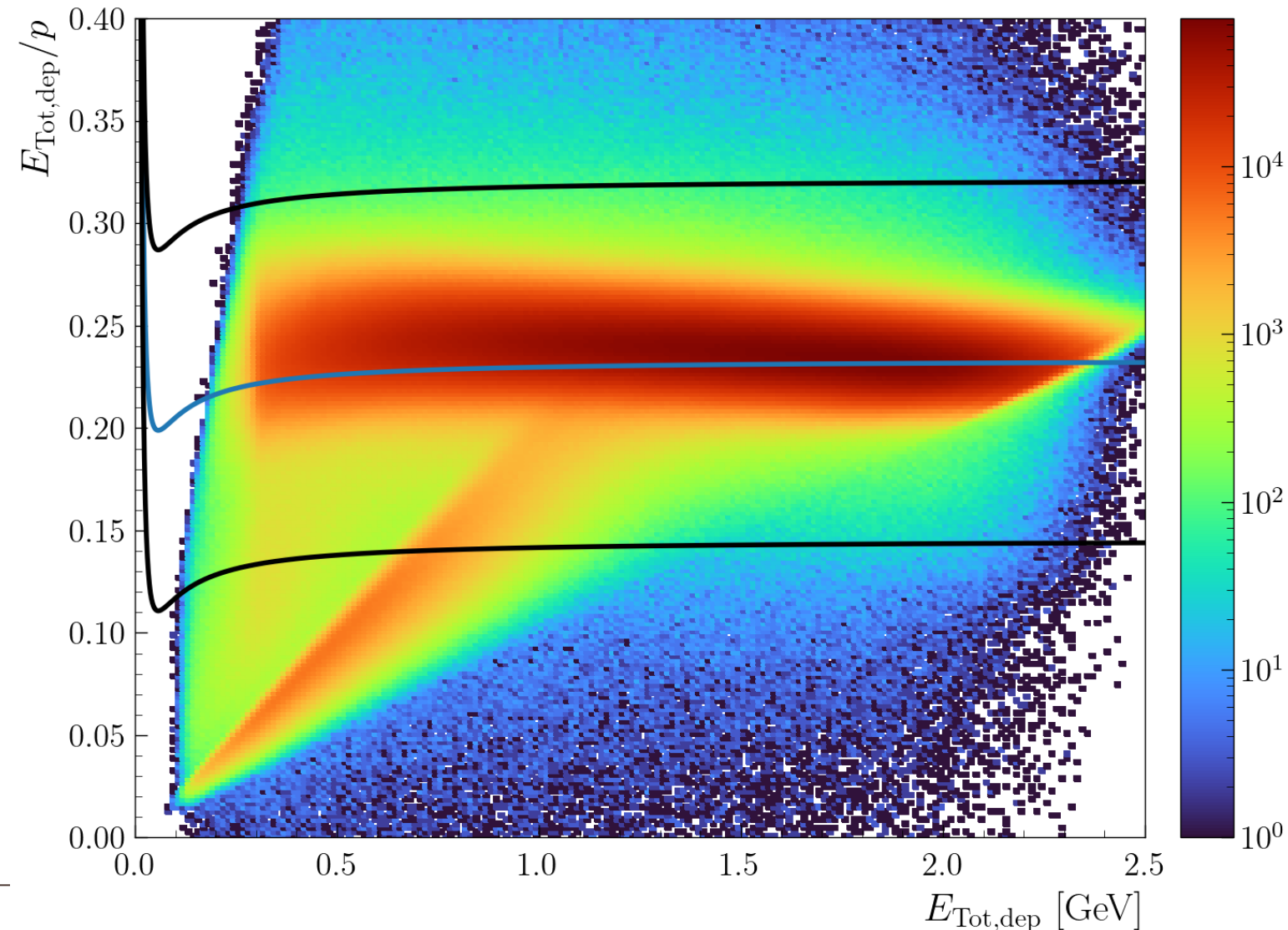
- The parameters a,b,c,d,e,f, and g can be found in the [CCDB](#).
- This cut is sector-dependent but target-independent.

RG-D: Electron identification

- Electron Particle ID with Event Builder: Sampling Fraction Cut

Target: LD2 - OB - Sector 1 - Pass Nphe + EPCal cuts

All negative tracks shown are with the $N_{\text{phe}} > 2$ and $E_{\text{PCal}} > 0.06$ GeV cuts applied. The blue line is the result of the fit to the mean, and the black lines are at $\mu \pm 5\sigma$.



RG-D: Electron identification

10

- Electron Identification Refinements: PCAL Minimum Energy Deposition Cut
 $E_{\text{PCal}} > 0.07 \text{ GeV}$

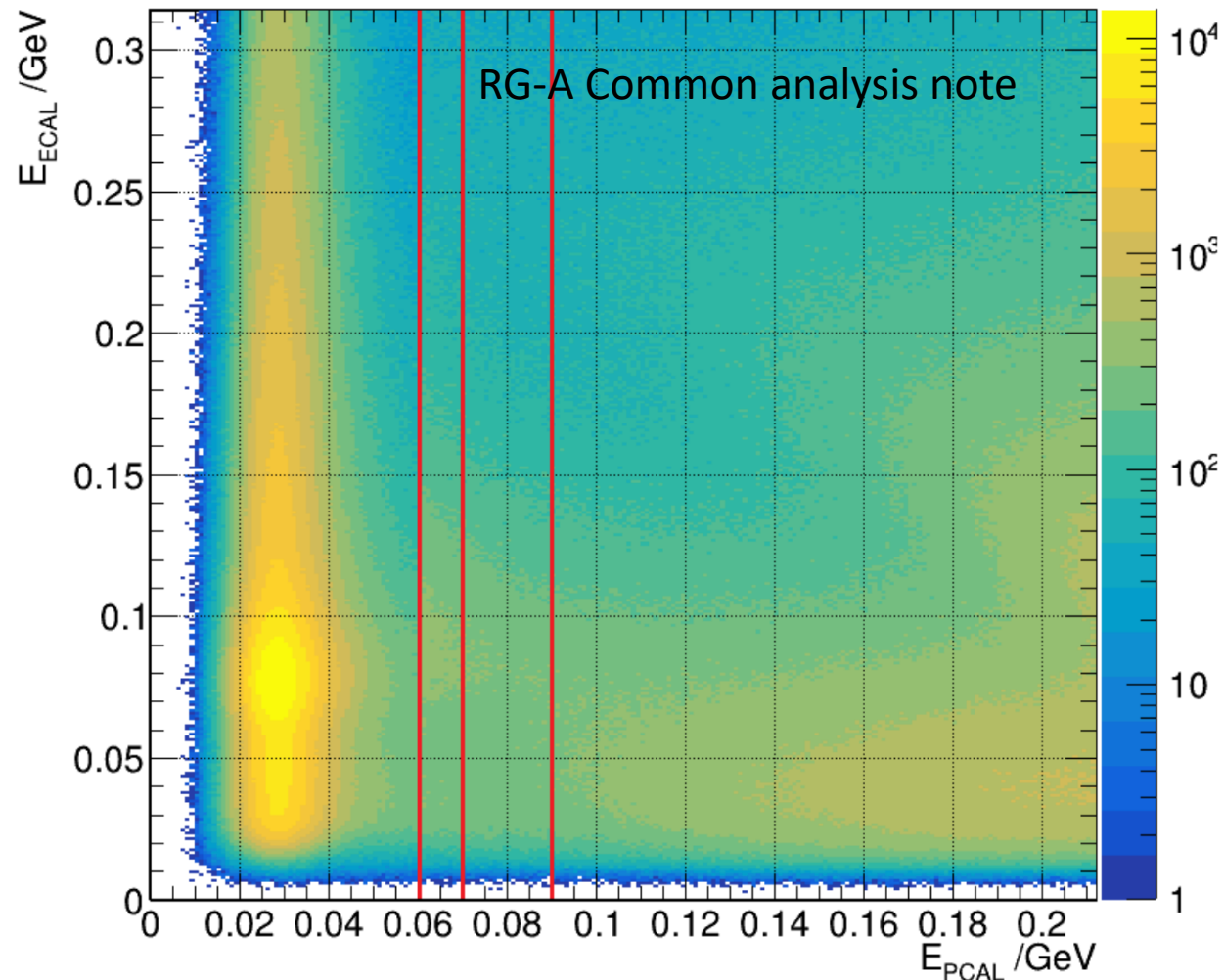
Purpose: Minimize negative pion contamination in electron samples

Pion distribution extends a bit further; a cut at 0.07 GeV is used. The rejection rate of these refinements is small. 99.85% of the Event Builder electron candidates survive the restriction of this cut from 0.06 to 0.07.

Only RG-A uses this cut.

RG-M/K doesn't use it.

RG-D doesn't need this cut.



RG-D: Electron identification

- Electron Identification Refinements: PCAL Minimum Energy Deposition Cut
 $E_{\text{PCal}} > 0.07 \text{ GeV}$

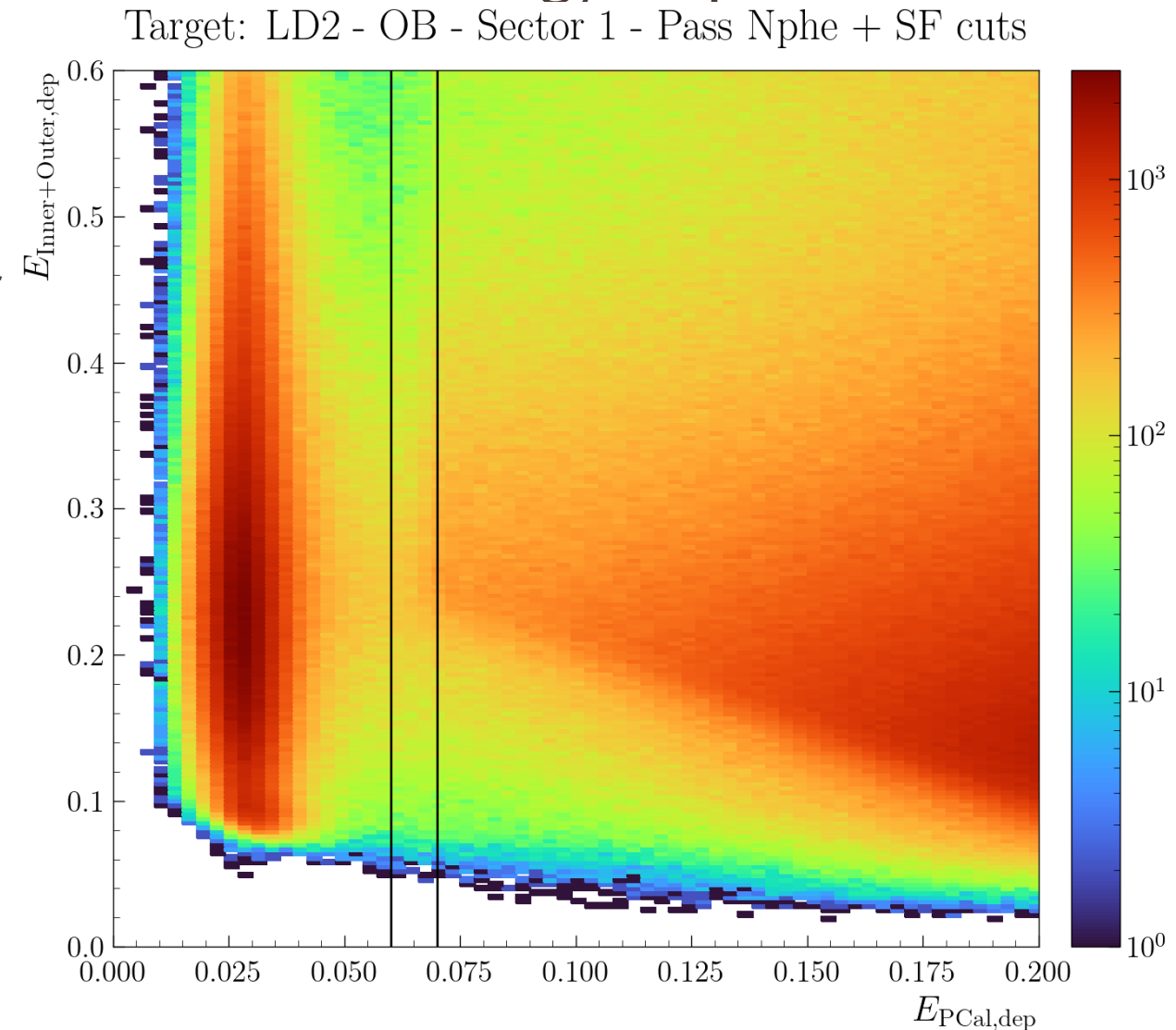
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Only RG-A uses this cut.

RG-M/K doesn't use it.

RG-D doesn't need this cut.



RG-D: Electron identification

• Electron Identification Refinements: Sampling Fraction vs momentum cut

- To improve the electron identification, another cut on the sampling fraction as a function of the momentum has been developed. For that, the SF is binned in momentum, and the resulting SF distribution is fit with a Gaussian distribution to extract the mean and sigma for a given momentum range. The momentum or energy dependence of the mean and sigma can then be fit to:

RG-M function

$$\mu_{SF}(x) = a + \frac{b}{x} + \frac{c}{x^2}$$

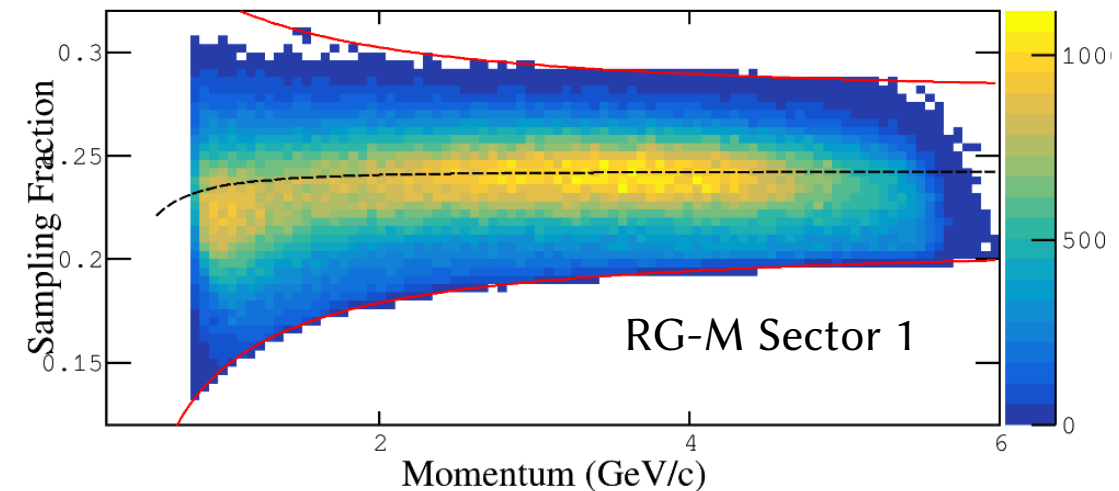
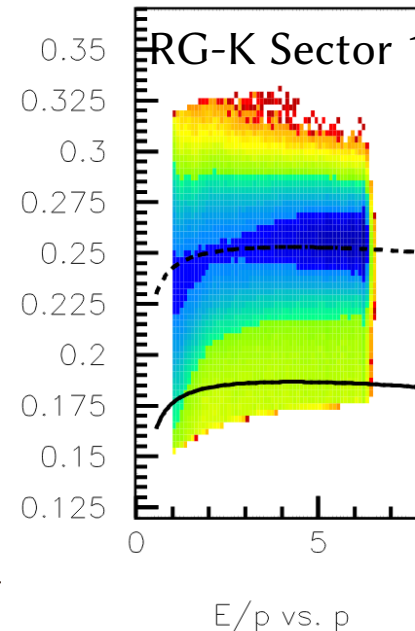
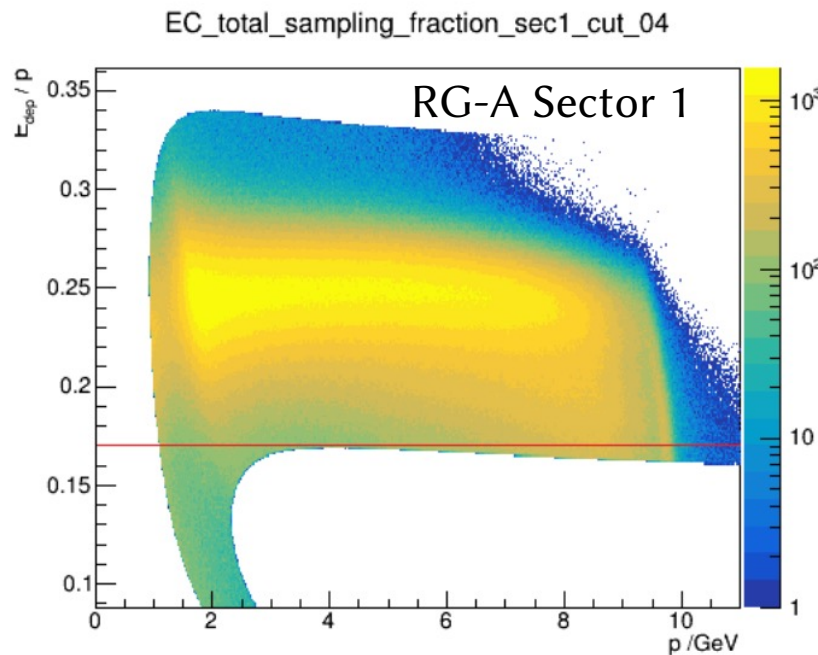
$$\sigma_{SF}(x) = a + \frac{b}{x} + \frac{c}{x^2}$$

RG-D function

$$\mu_{SF} = a_{\mu} + b_{\mu}p + c_{\mu}p^2 + d_{\mu}p^3$$

$$\sigma_{SF} = a_{\sigma} + b_{\sigma}p + c_{\sigma}p^2 + d_{\sigma}p^3$$

- RG-A/K: I don't know which function they use.

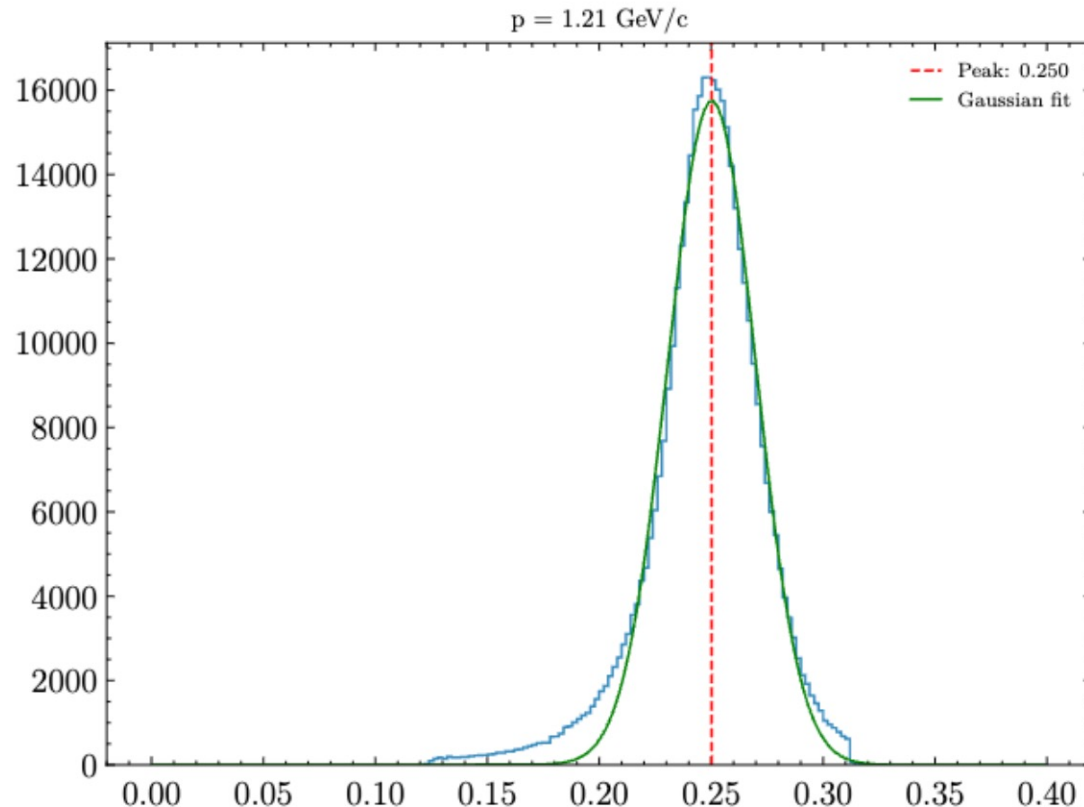


RG-D: Electron identification

- Electron Identification Refinements: Sampling Fraction Cut vs momentum $\mu \pm 3\sigma$
 - To improve the electron identification, another cut on the sampling fraction as a function of the momentum has been developed. For that, the SF is binned in momentum, and the resulting SF distribution is fit with a Gaussian distribution to extract the mean and sigma for a given momentum range. The momentum or energy dependence of the mean and sigma can then be fit to:

$$\mu_{\text{SF}} = a_{\mu} + b_{\mu}p + c_{\mu}p^2 + d_{\mu}p^3$$

$$\sigma_{\text{SF}} = a_{\sigma} + b_{\sigma}p + c_{\sigma}p^2 + d_{\sigma}p^3$$



Target LD2, OB Sector 1
Slice for bin center: 1.21 GeV/c

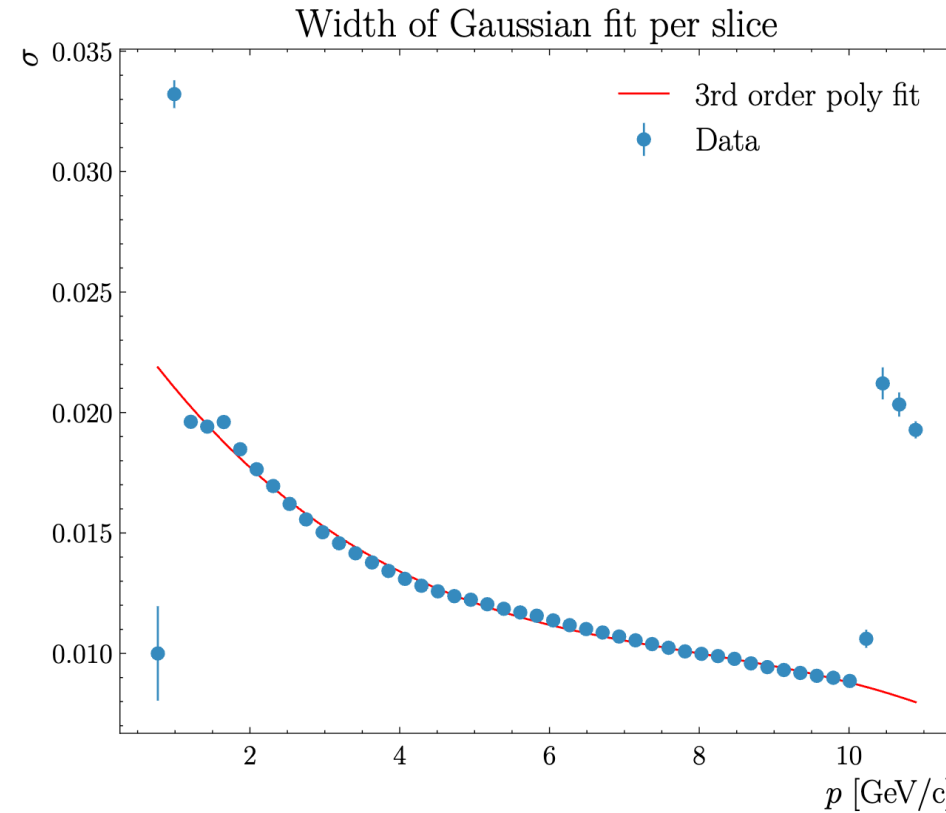
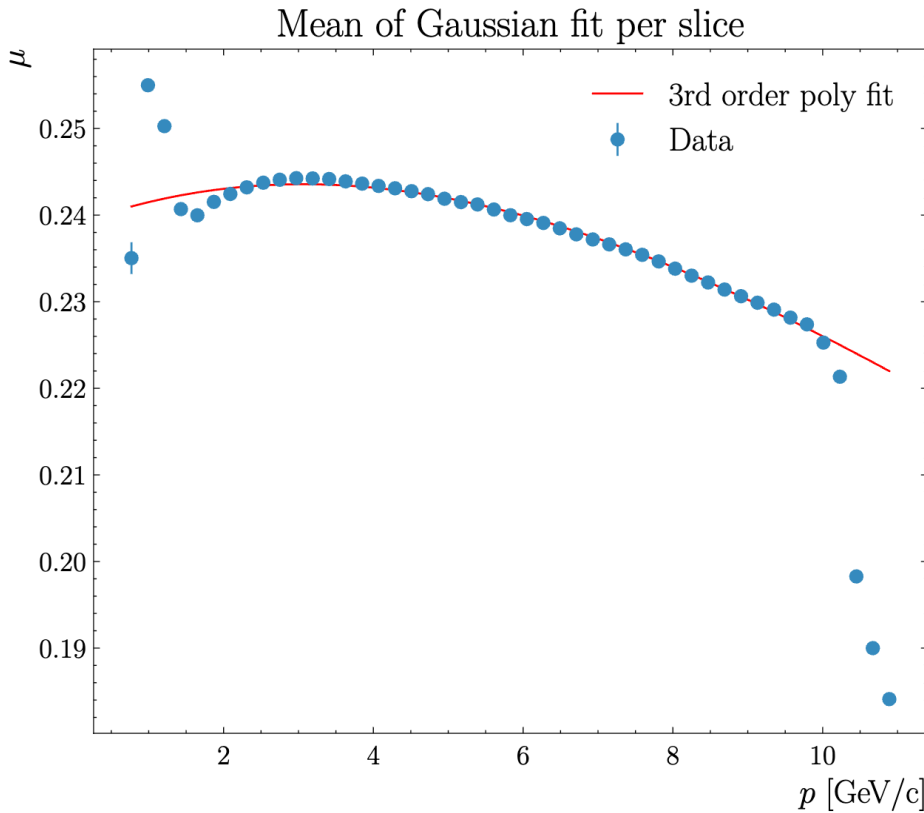
RG-D: Electron identification

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$$\mu_{\text{SF}} = a_{\mu} + b_{\mu}p + c_{\mu}p^2 + d_{\mu}p^3$$

$$\sigma_{\text{SF}} = a_{\sigma} + b_{\sigma}p + c_{\sigma}p^2 + d_{\sigma}p^3$$

Target LD2, OB Sector 1

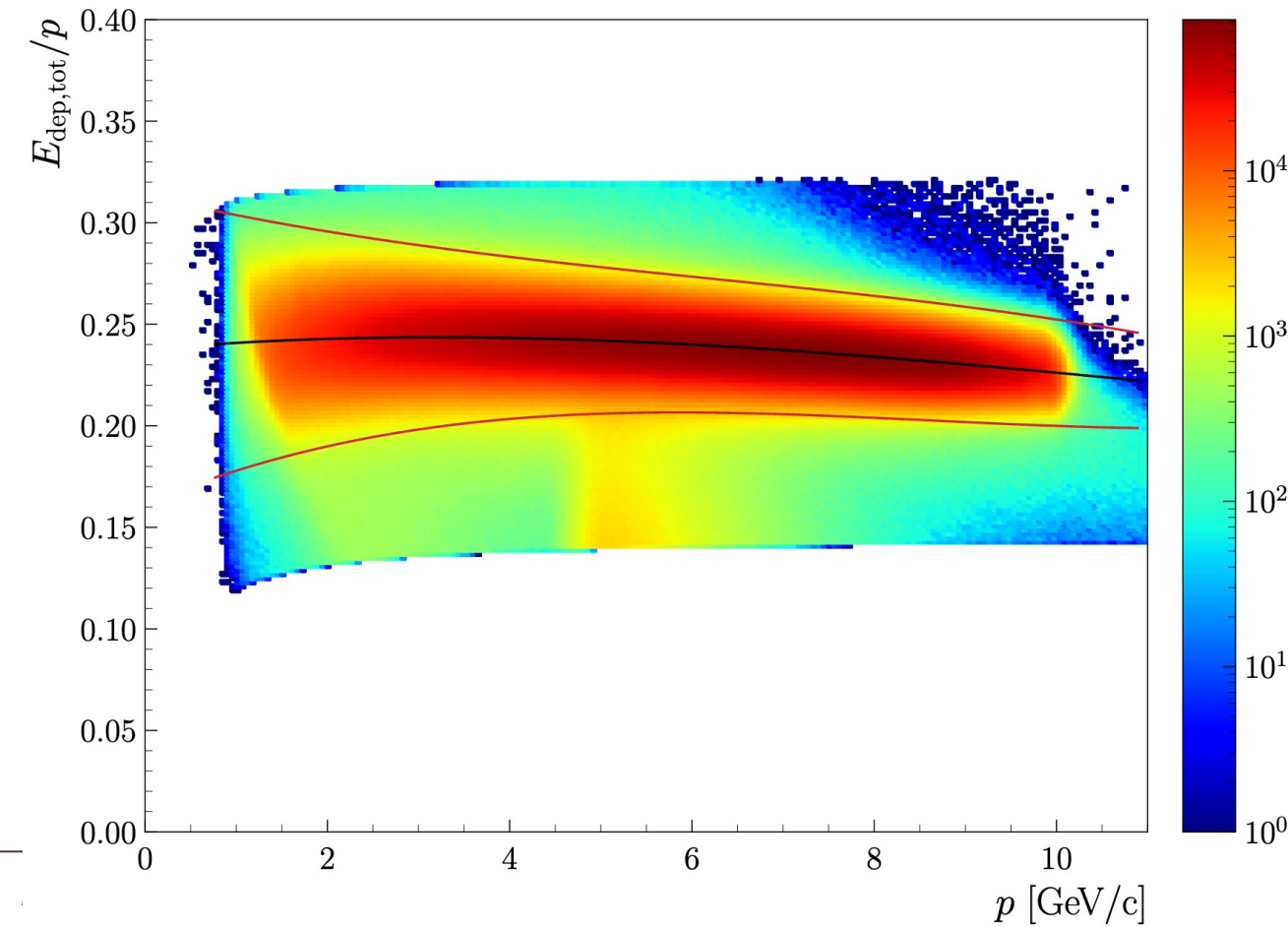


RG-D: Electron identification

- Electron Identification Refinements: Sampling Fraction Cut vs momentum $\mu \pm 3\sigma$
 - To improve the electron identification, another cut on the sampling fraction as a function of the momentum has been developed. For that, the SF is binned in momentum, and the resulting SF distribution is fit with a Gaussian distribution to extract the mean and sigma for a given momentum range. The momentum or energy dependence of the mean and sigma can then be fit to:

$$\mu_{\text{SF}} = a_{\mu} + b_{\mu}p + c_{\mu}p^2 + d_{\mu}p^3$$

$$\sigma_{\text{SF}} = a_{\sigma} + b_{\sigma}p + c_{\sigma}p^2 + d_{\sigma}p^3$$



Target LD2, OB Sector 1

This cut is sector, target, and torus polarity dependent

Use by RG-A/K/M

RG-D uses this cut, too.

RG-D: Electron identification

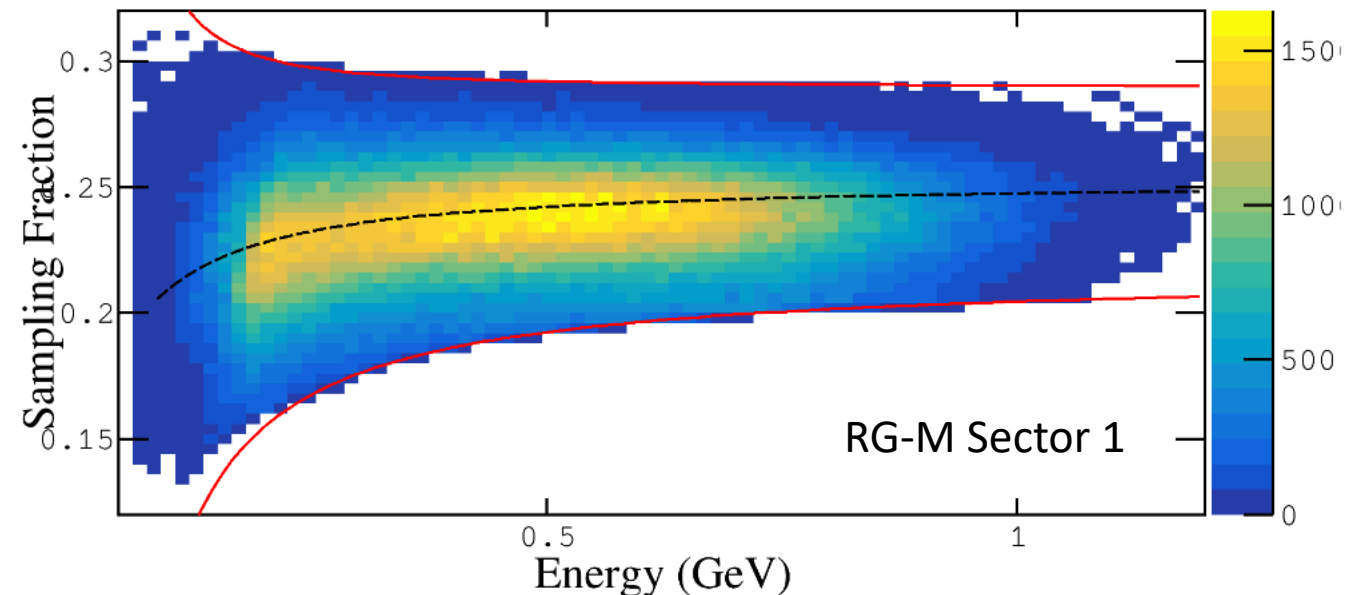
- Electron Identification Refinements: Sampling Fraction Cut vs Total Energy deposited $\mu \pm 3\sigma$
- Same strategy as the momentum cut
- Only RG-M uses this cut
- It is also a cut: $-3 < \text{chi2pid} < 3$

RG-M Fit function:

$$\mu_{SF}(x) = a + \frac{b}{x} + \frac{c}{x^2}$$

$$\sigma_{SF}(x) = a + \frac{b}{x} + \frac{c}{x^2}.$$

RG-A, RG-K, and RG-D use default Sampling Fraction vs Edep $\mu \pm 5\sigma$

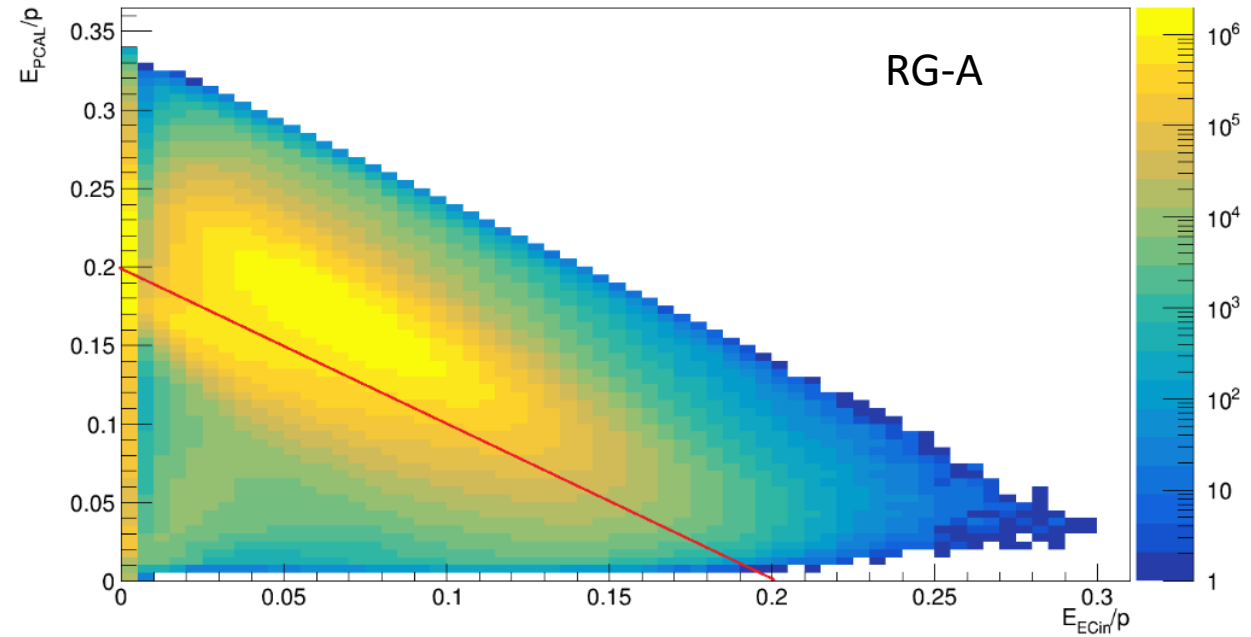


RG-D: Electron identification

- Electron Identification Refinements: Cut on the sampling fraction of the inner calorimeter and the PCAL

Detailed contamination studies showed that above the HTCC threshold at 4.5 GeV, a significant pion contamination is present.

Therefore, another cut has been placed on the correlation between the sampling fractions of the inner calorimeter and the PCAL

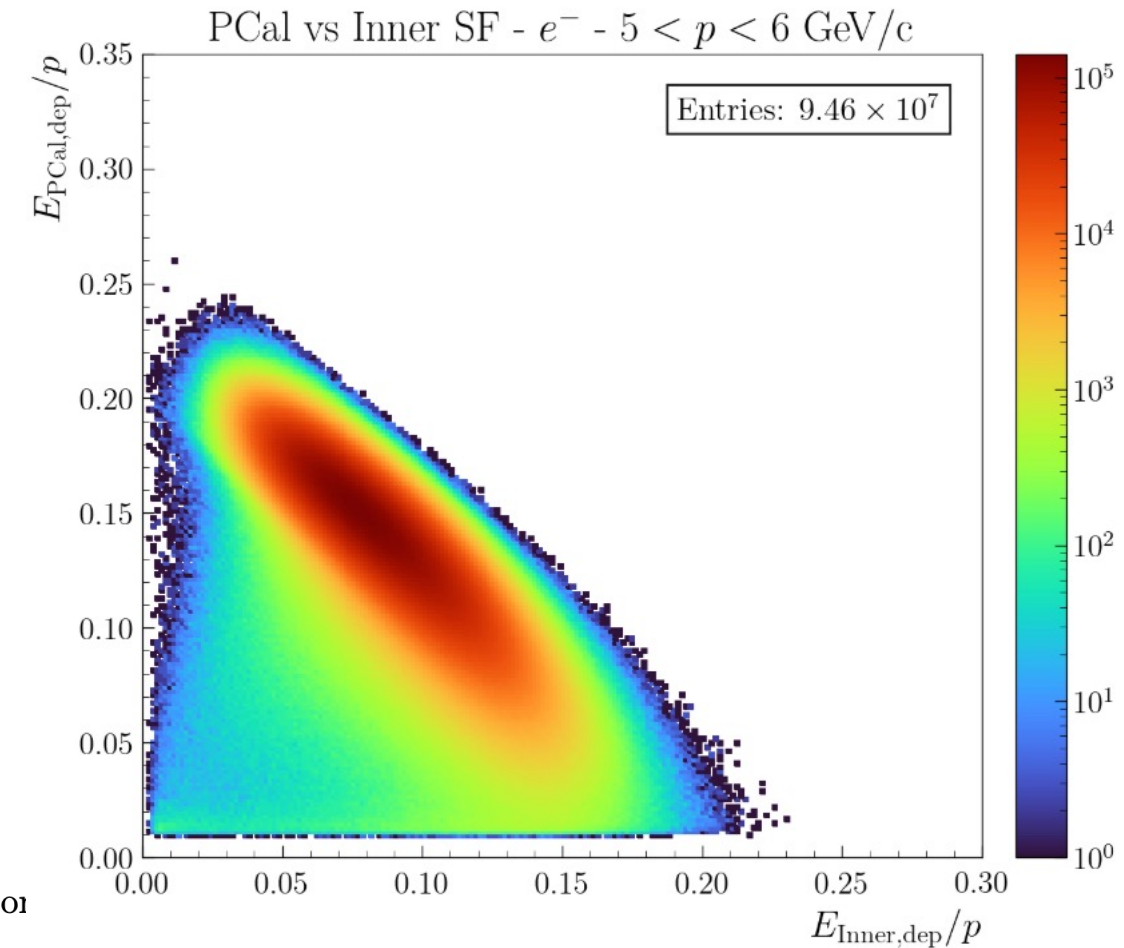


RG-D: Electron identification

- Electron Identification Refinements: Cut on the sampling fraction of the inner calorimeter and the PCAL

After applying the SF vs p and fiducial cuts for RG-D the correlation doesn't have contamination for the outbending Target LD2, outbending

This cut is using by RG-A/K
RG-M doesn't use it
RG-D doesn't need it



9. Page 14, eq. 2 - there is a confusion about what sampling fraction is used. The equation total, but the analysis only seems to use $(E_{PCAL} + E_{ecin})/P$.

RG-M note

Sampling fraction cuts are calculated as the total sum of all calorimeters. There is an extra cut which was used in RGA analysis note which we copied where they cut on the correlation between the PCAL and ECIN sampling fraction cuts. After discussion with the reviewer we decided that it is not a cut to be used in this analysis and it is not well motivated, so we removed it.

RG-D: Electron identification

- Electron Identification Refinements: Cut on the electron momentum
- RG-M Common analysis note comment:

1070 6. **Electron momentum cut**, I see from Figs. 5 and 7 that you consider all the electrons without
1071 any cut on the momentum. In CLAS12 we don't trust the electron reconstruction below 800MeV/c.
1072 Just for reference, RG-B cuts the electrons below 1 GeV/c. So please apply a minimum momentum
1073 cut before showing any cut later on.

1074

1075 A min momentum cut of 800 MeV/c was placed now for all electrons.

- We should probably apply a 0.8 GeV cut on the momentum of the electron

RG-D: Electron identification

- Electron Identification Refinements: Vertex cuts
- RG-A/M and K use just a Vz cut (no cut on Vx and Vy):
- LD2:
 - $vz_min_e = -15 \text{ cm}$
 - $vz_max_e = 5;$
- CxC:
 - $vz_min_e = -10.6;$
 - $vz_max_e = 5;$
- Cu:
 - $vz_min_e = -10.6;$
 - $vz_max_e = -6.5;$
- Sn:
 - $vz_min_e = -5.5;$
 - $vz_max_e = 5;$

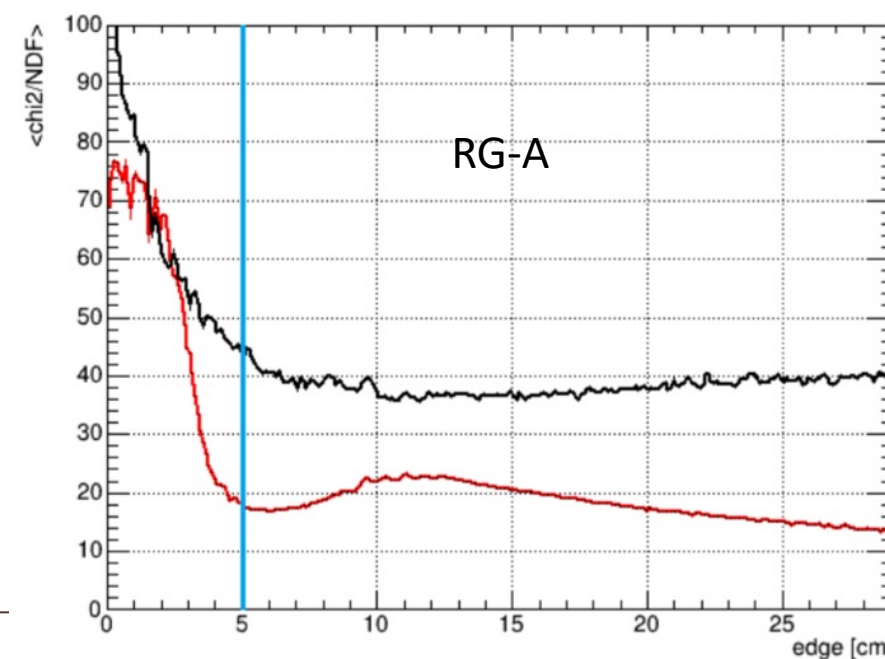
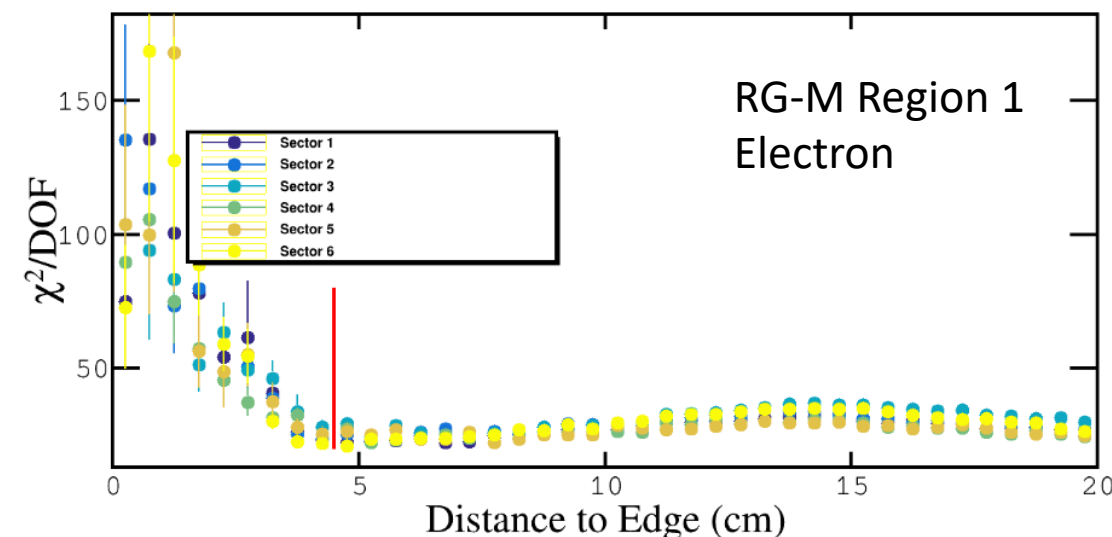
RG-D: Electron identification

• Electron DC Fiducial cuts:

- RG-A (pass2) and RG-M use χ^2/ndf vs edge
- RG-K uses the old χ^2/ndf in the xy-plane
- RG-D uses θ_{DC} vs edge
- RG-C seems to also use χ^2/ndf vs edge
- RG-M has different cuts for electron and proton:
 - 4.5, 3.5, 7.5 cm for electrons and 2.5, 3.0, 10.5 cm
 - for protons for regions 1,2, and 3

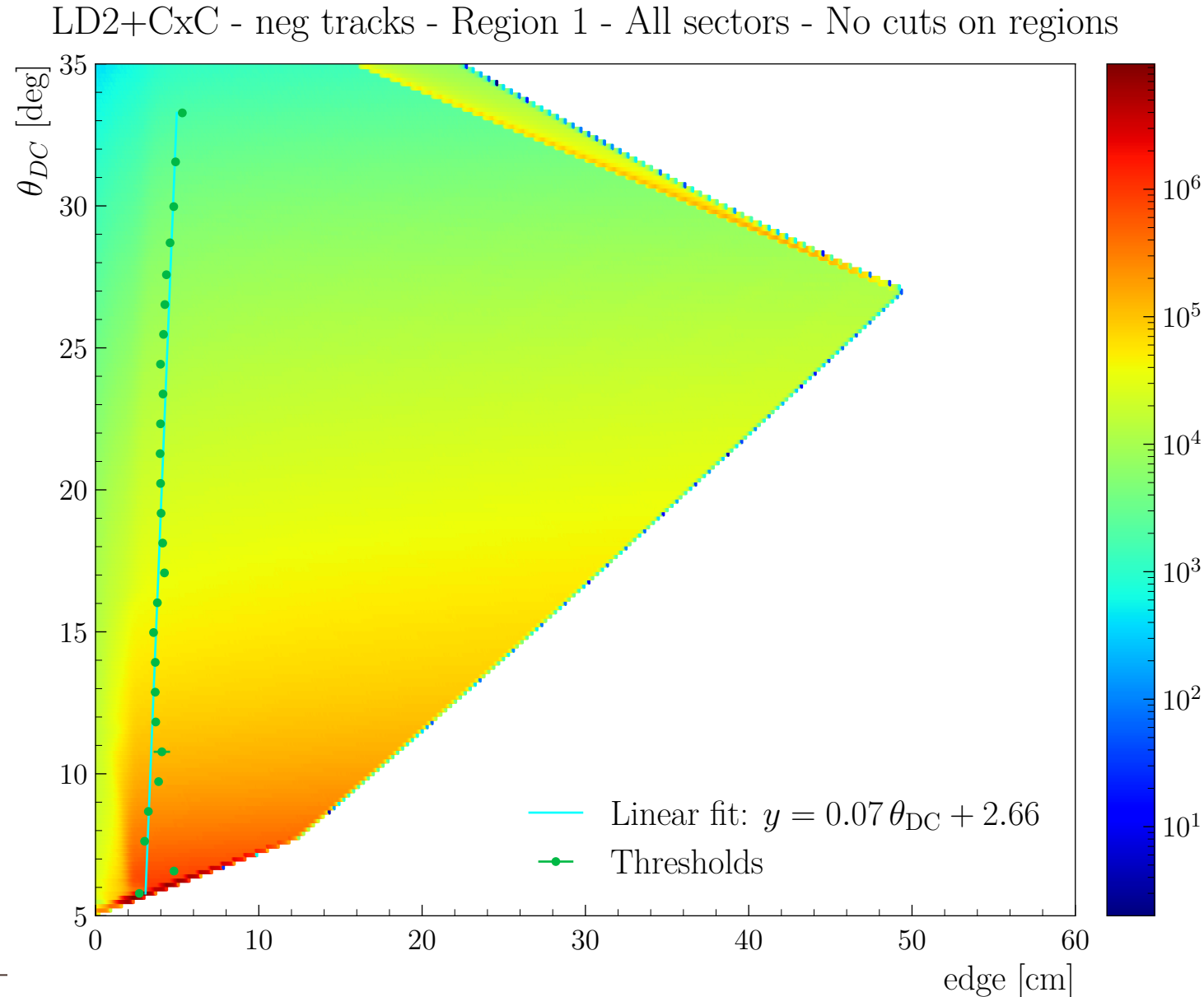
○ RG-A:

particle	field	region 1	region 2	region 3
electrons	in-bending	5.0	5.0	10.0
protons	in-bending	2.5	2.5	9.0
π^+	in-bending	2.5	2.5	9.0
π^-	in-bending	3.5	3.0	7.0
K^+	in-bending	2.5	2.0	9.0
K^-	in-bending	3.5	2.5	5.0
electrons	out-bending	3.0	3.0	10.0
protons	out-bending	3.5	3.0	7.0
π^+	out-bending	3.5	2.5	6.5
π^-	out-bending	2.5	2.5	10.0
K^+	out-bending	3.5	2.5	6.5
K^-	out-bending	2.5	2.5	10.0



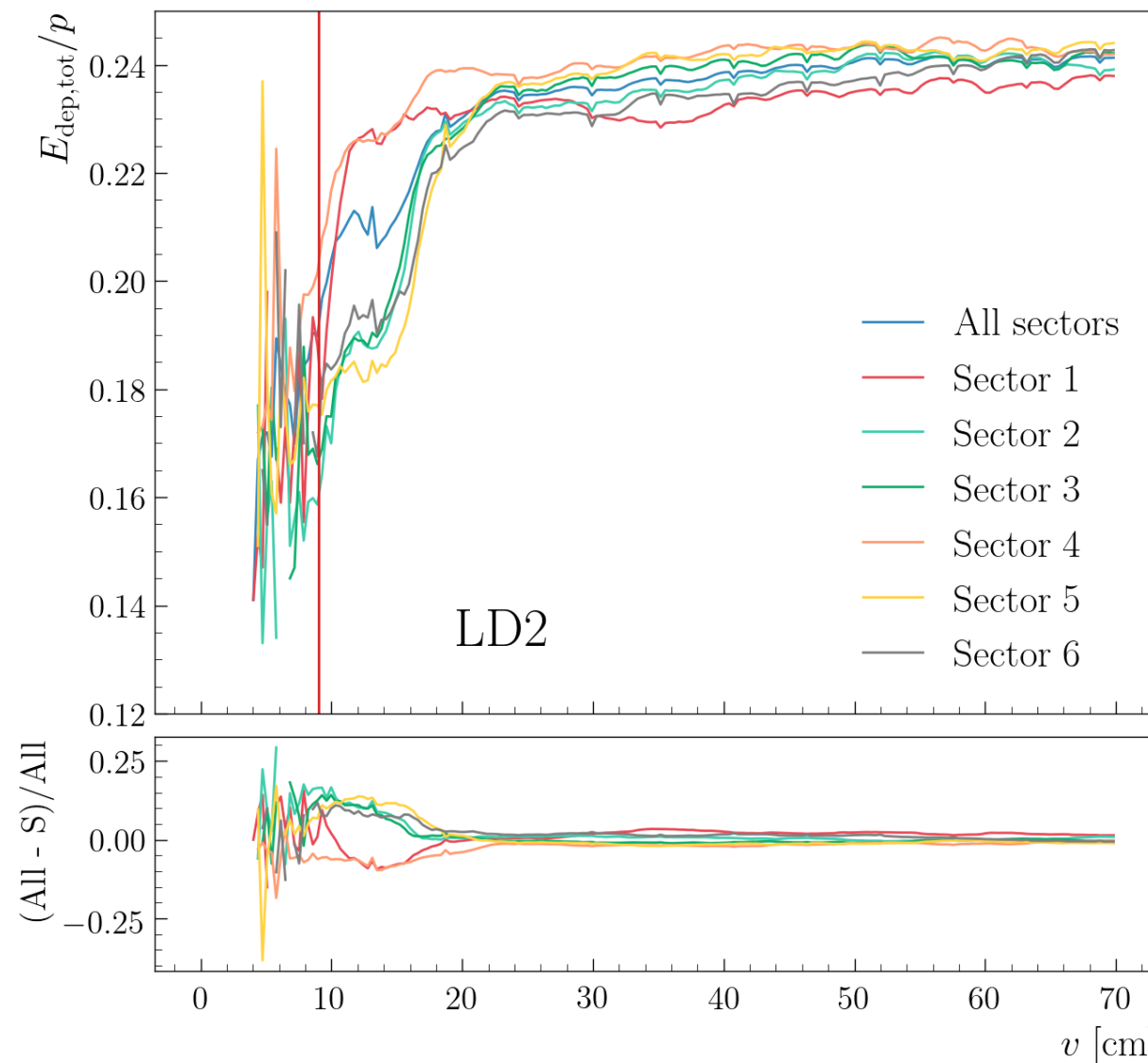
Backup: DC edge

- Cut on the DC edge variation as a function of the local θ in the DC plan.
- One cut for each DC region:
 - Region 1: $0.07 \theta_{\text{DC}} + 2.66 \text{ cm}$
 - Region 2: $0.02 \theta_{\text{DC}} + 2.80 \text{ cm}$
 - Region 3: $0.30 \theta_{\text{DC}} + 3.27 \text{ cm}$



RG-D: Electron identification

- Electron ECal Fiducial cuts:
 - V and $W > 9$ cm for the PCal



RG-D: Electron identification

- Summary: Electron Identification Refinements
 - SF vs p: `/work/clas12/ouillon/RGD/rgd_fiducial/include/Core/SamplingFractionCut.hpp`
 - DC Fiducial cuts: R1: $0.07 \theta_{\text{DC}} + 2.66 \text{ cm}$ R2: $0.02 \theta_{\text{DC}} + 2.80 \text{ cm}$ R3: $0.30 \theta_{\text{DC}} + 3.27 \text{ cm}$
 - ECal Fiducial cut: V and W > 9 cm
 - Momentum: $p > 0.8 \text{ GeV}$
 - Vz vertex:

 - Survival rate LD2:
 - | Total events: 16228266 |
 - | With electron: 10505730 (100.00%) |
 - | SF: 6527626 (62.13%) |
 - | +edges: 5318510 (50.62%) |
 - | +v/w: 5318508 (50.62%) |
 - | +p>0.8 GeV: 5318507 (50.62%) |
 - | +vz cut: 5135739 (48.89%) |

Pass1 status

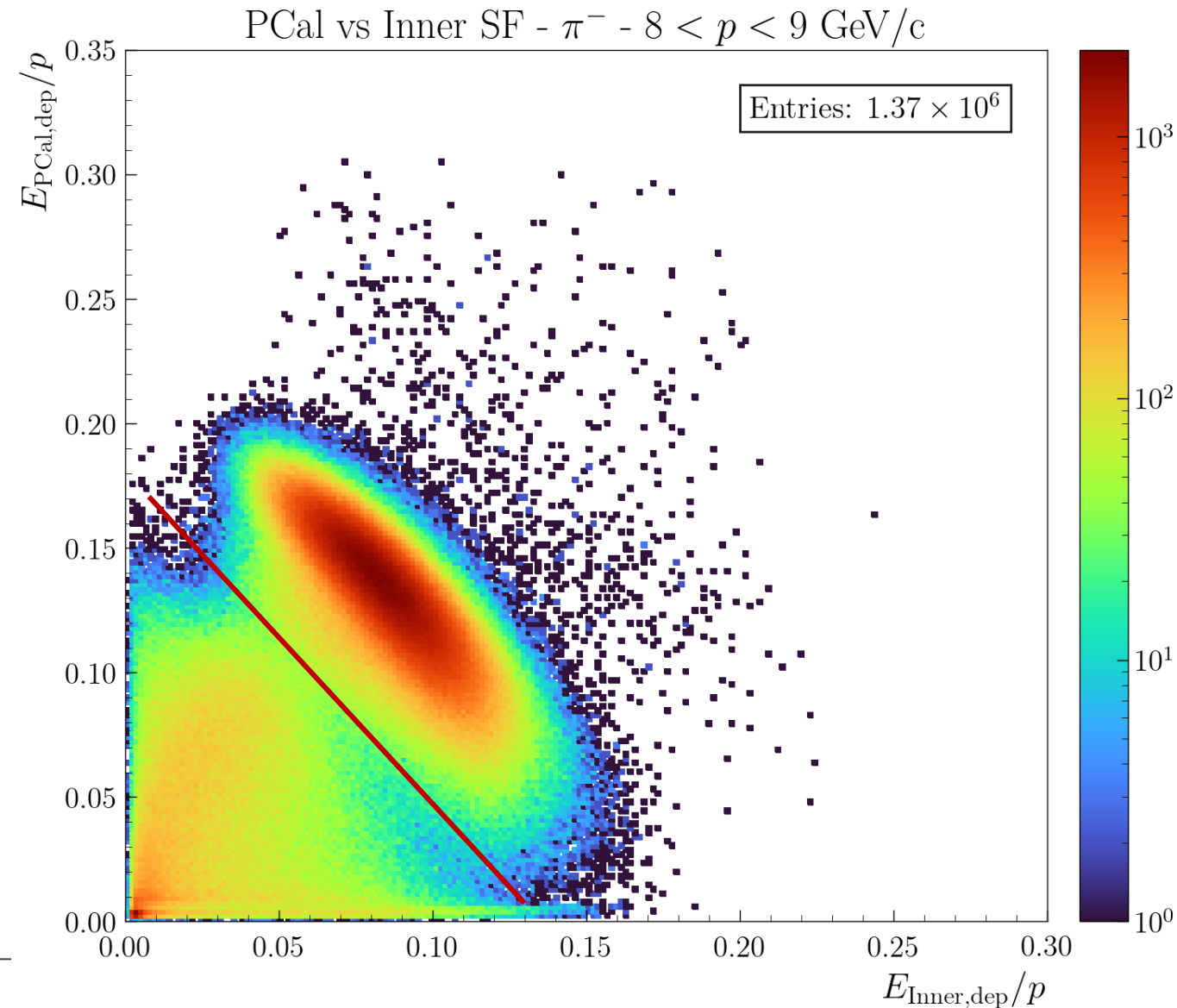
Workflow Name	Jobs	Succeeded	Dispatched	Problems	Attempts	Pending	Preparing	Running	Reaping
rgd-dmra-pass1_CuSn_1-18561x33	14687	14658	22	0	19811	22	0	0	0
rgd-dmra-pass1_CuSn_2-18347x294	70791	5000	5000	0	15000	5000	0	0	0
rgd-dmra-pass1_CxC_1-18445x13	6873	6873	0	0	6890	0	0	0	0
rgd-dmra-pass1_CxC_2-18339x116	36705	13863	5000	0	23907	5000	0	0	0
rgd-dmra-pass1_LD2_1-18431x16	7702	7702	0	0	7729	0	0	0	0
rgd-dmra-pass1_LD2_2-18305x144	40076	0	5000	5000	15000	0	0	0	0
rgd-dmra-pass1_empty_nominal-183...	113	113	0	0	185	0	0	0	0
rgd-dmra-pass1_empty_zero_field-...	239	239	0	0	394	0	0	0	0
TOTALS	177186	48448	15022	5000	88916	10022	0	0	0
SUMMARY STATISTICS									
Total Jobs: 177,186 Succeeded: 48,448 Running: 15,022 Pending: 113,716 Progress Percentages: Completed: 27.34% Running: 8.48% Pending: 64.18% Progress Bar: Done Running Pending									

You can access the Python script here:

https://gitlab.com/MathieuOuillon/rgd_workflow_stats

RG-D: π^- SF Inner vs SF PCal

- I am still working on the SF of the Inner part vs SF of the pre-shower.
- This band above the red line:
 - Data LD2 Pass1 OB
 - cut: pid = -211
 - cut on chi2pid vs p doesn't improve
 -

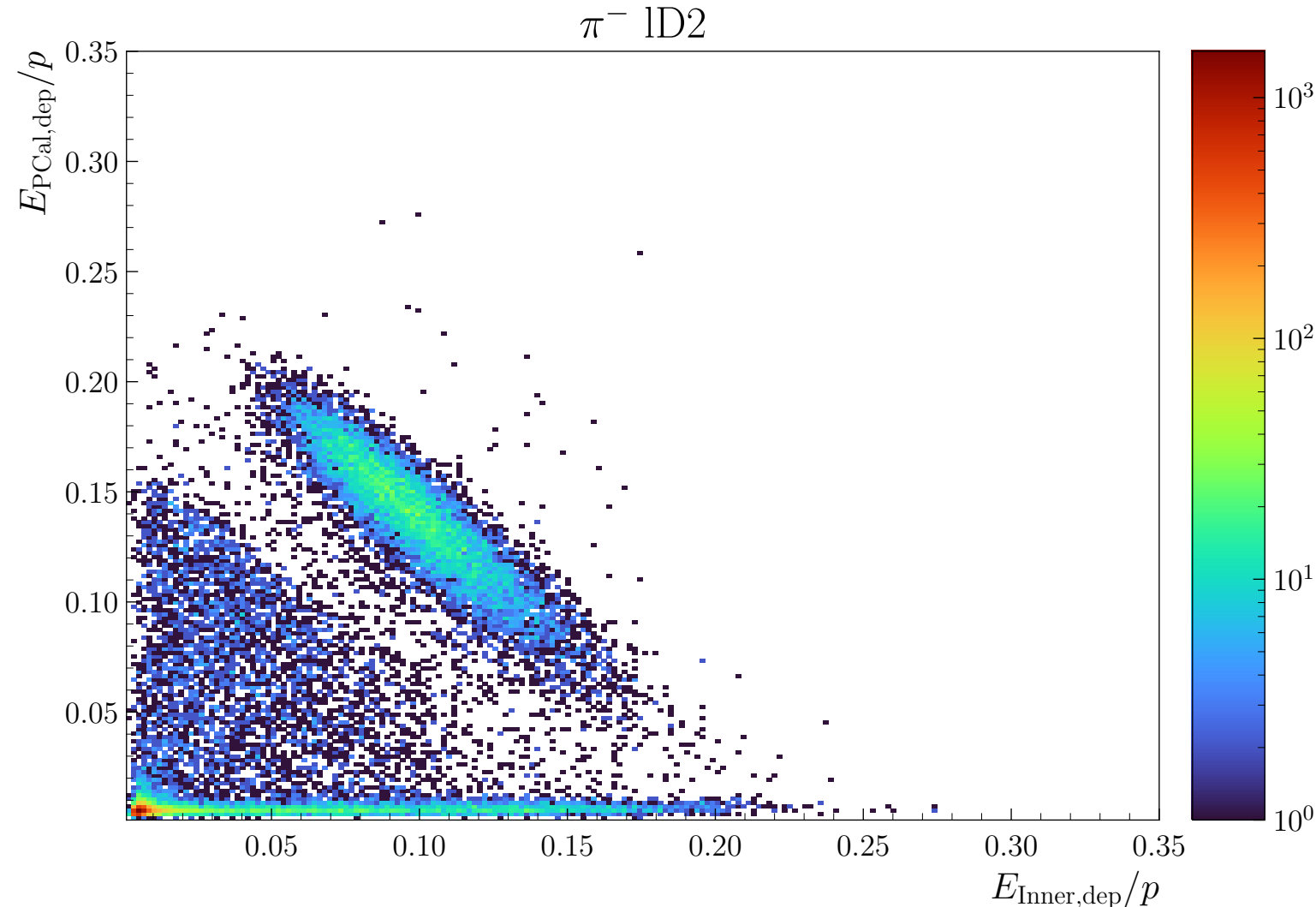


RG-D: π^- SF Inner vs SF PCal

- I am still working on the SF of the Inner part vs SF of the pre-shower.

Use simulation to identify particles in that band. $5 < p_{\pi^-} < 12$ GeV

Here it is all particles reconstructed as π^- (pid=-211)



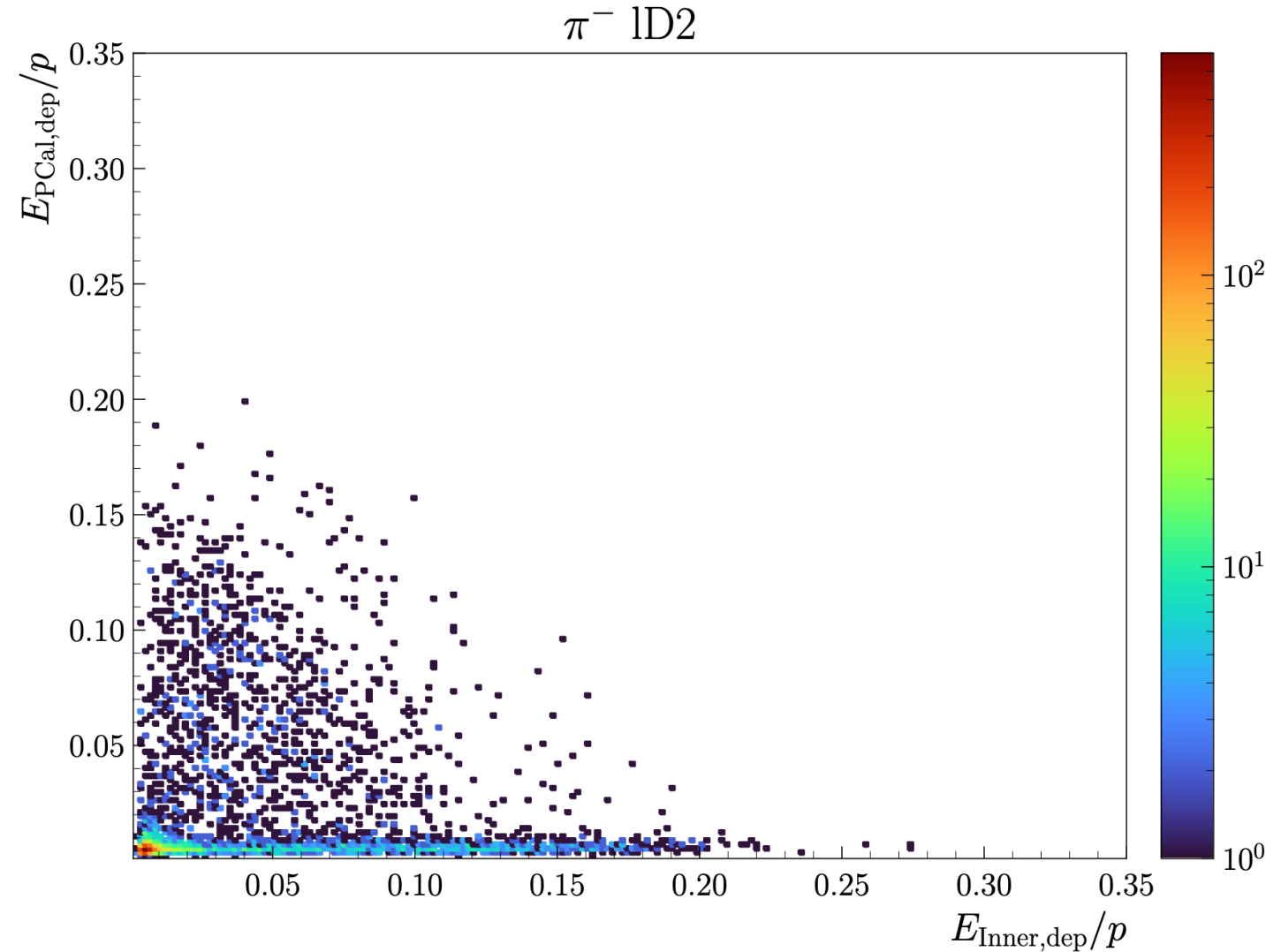
RG-D: π^- SF Inner vs SF PCal

- I am still working on the SF of the Inner part vs SF of the pre-shower.

Use simulation to identify particles in that band. $5 < p_{\pi^-} < 12$ GeV

Here it is all particles reconstructed as π^- (pid=-211) that match a generated π^-

Matched: momentum reconstructed is close to the generated (difference in p_x , p_y and p_z is less than 10%)



RG-D: π^- SF Inner vs SF PCal

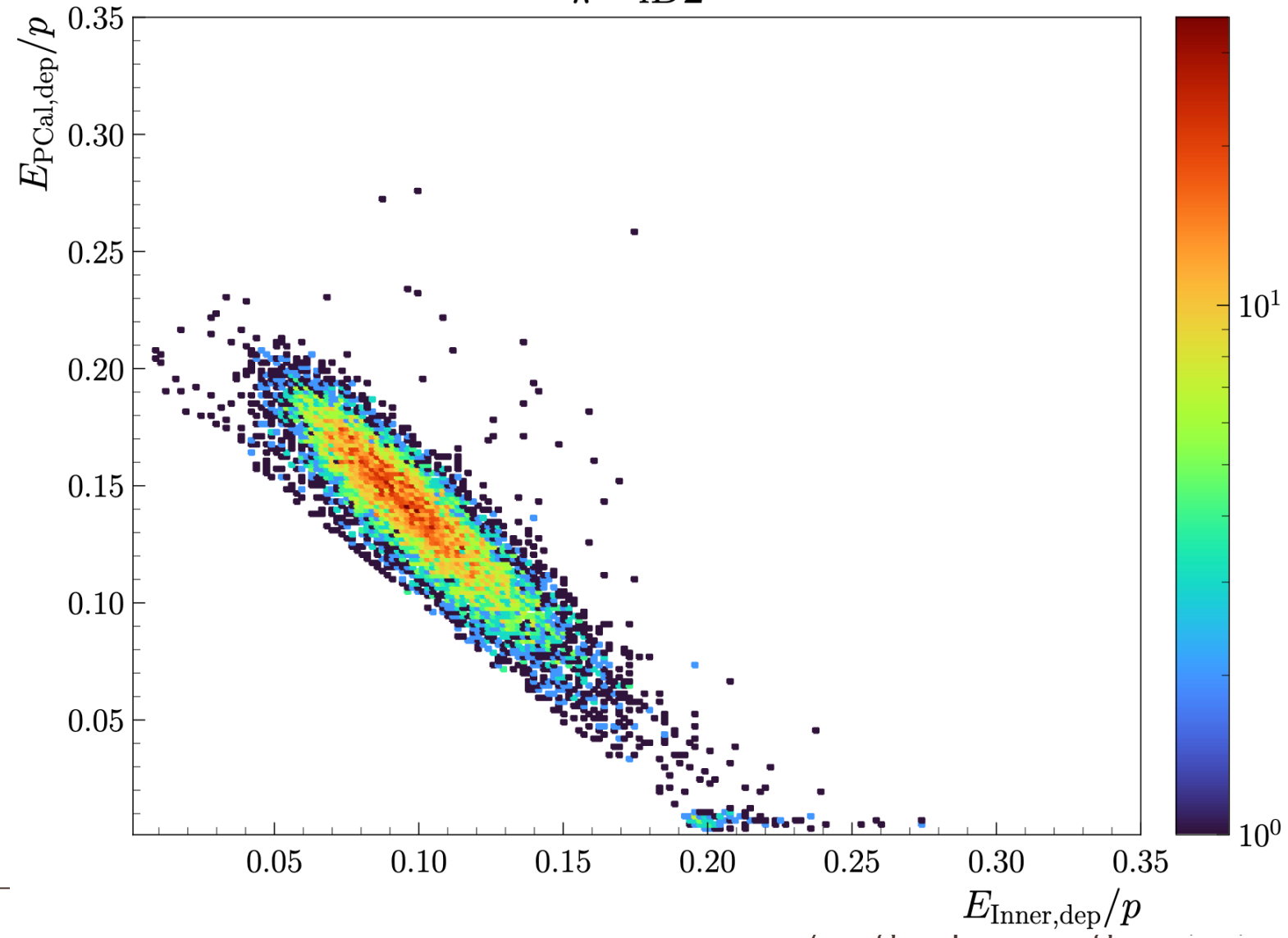
- I am still working on the SF of the Inner part vs SF of the pre-shower.

π^- 1D2

$5 < p_{\pi^-} < 12$ GeV

If I select the band with the cut:

$SF_{\text{Inner}} > 0.2 - SF_{\text{PCal}}$



RG-D: π^- SF Inner vs SF PCal

- I am still working on the SF of the Inner part vs SF of the pre-shower.

$$5 < p_{\pi^-} < 12 \text{ GeV}$$

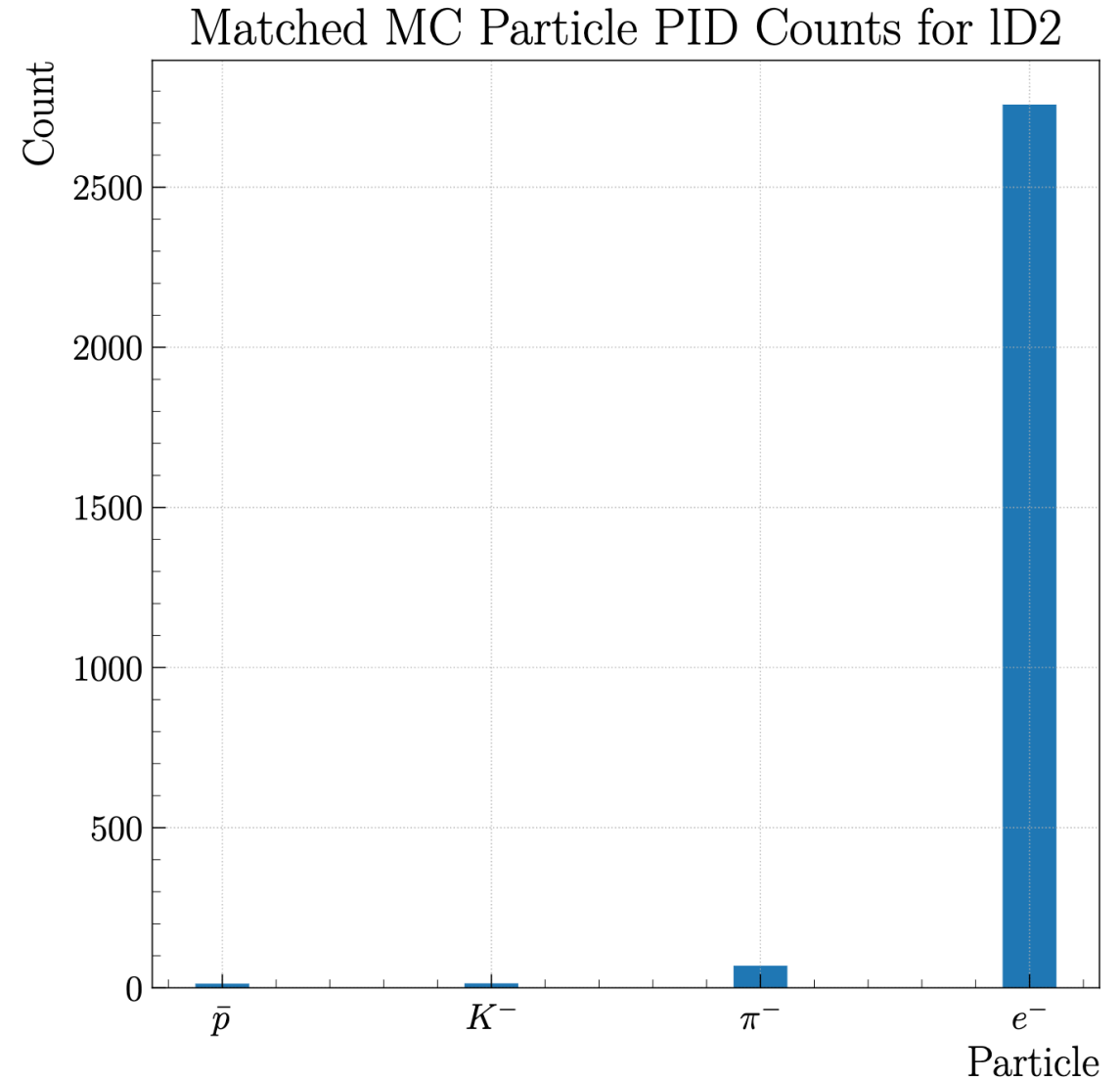
If I select the band with the cut:

$$\text{SF}_{\text{Inner}} > 0.15 - \text{SF}_{\text{PCal}}$$

And look at the generated particles that matched the reconstructed particles in the band

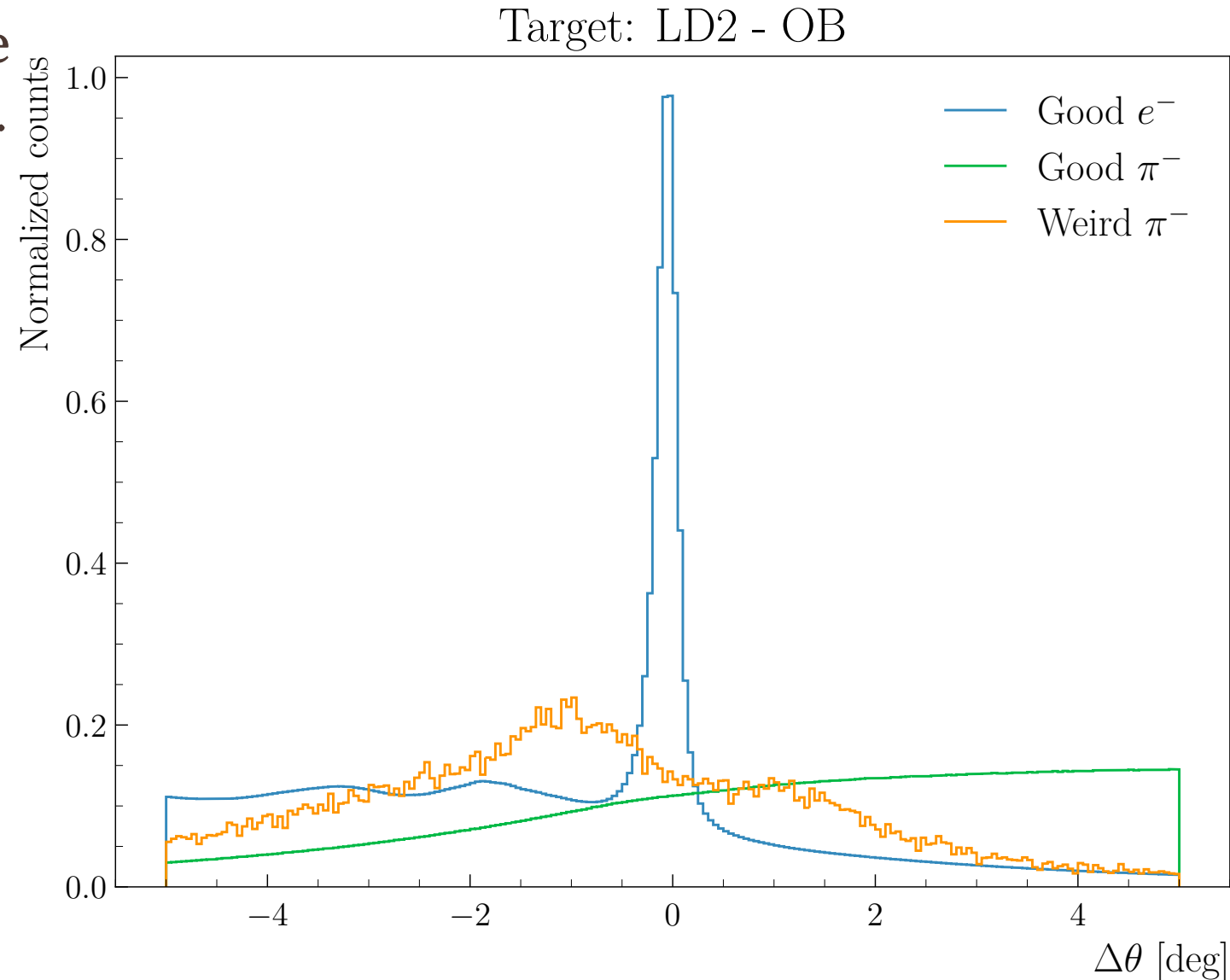
Looks like in the simulation, most of the particles in the band are electrons.

3000 electrons over 5M events



RG-D: π^- SF Inner vs SF PCal

- I am still working on the SF of the Inner part vs SF of the pre-shower.
- One more test: Electron radiates some photons, but not π^- , so by looking at the number of neutral particles close in $\Delta\theta$, we can know if it is an electron or a pion:

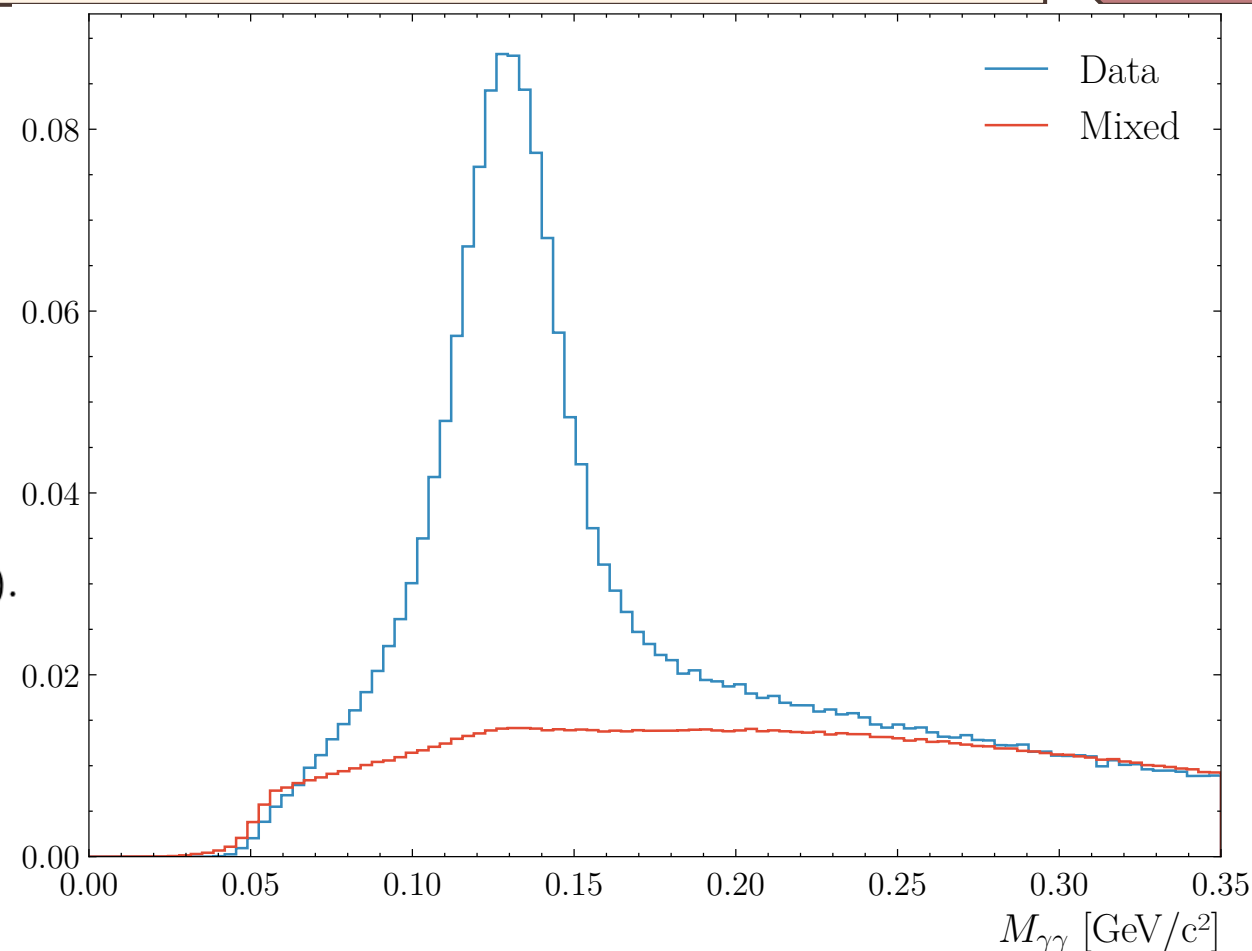


RG-D: π^0 Analysis

- Plan:
- Implement the event mixing
- Compare the 2-photon mass distribution for the different targets
- Extract multidimensional multiplicity ratio:

two sets of four-fold bins: $(Q^2, x_B, \nu$ and $z_h)$ and $(x_B, \nu, z_h$ and $p_T)$.

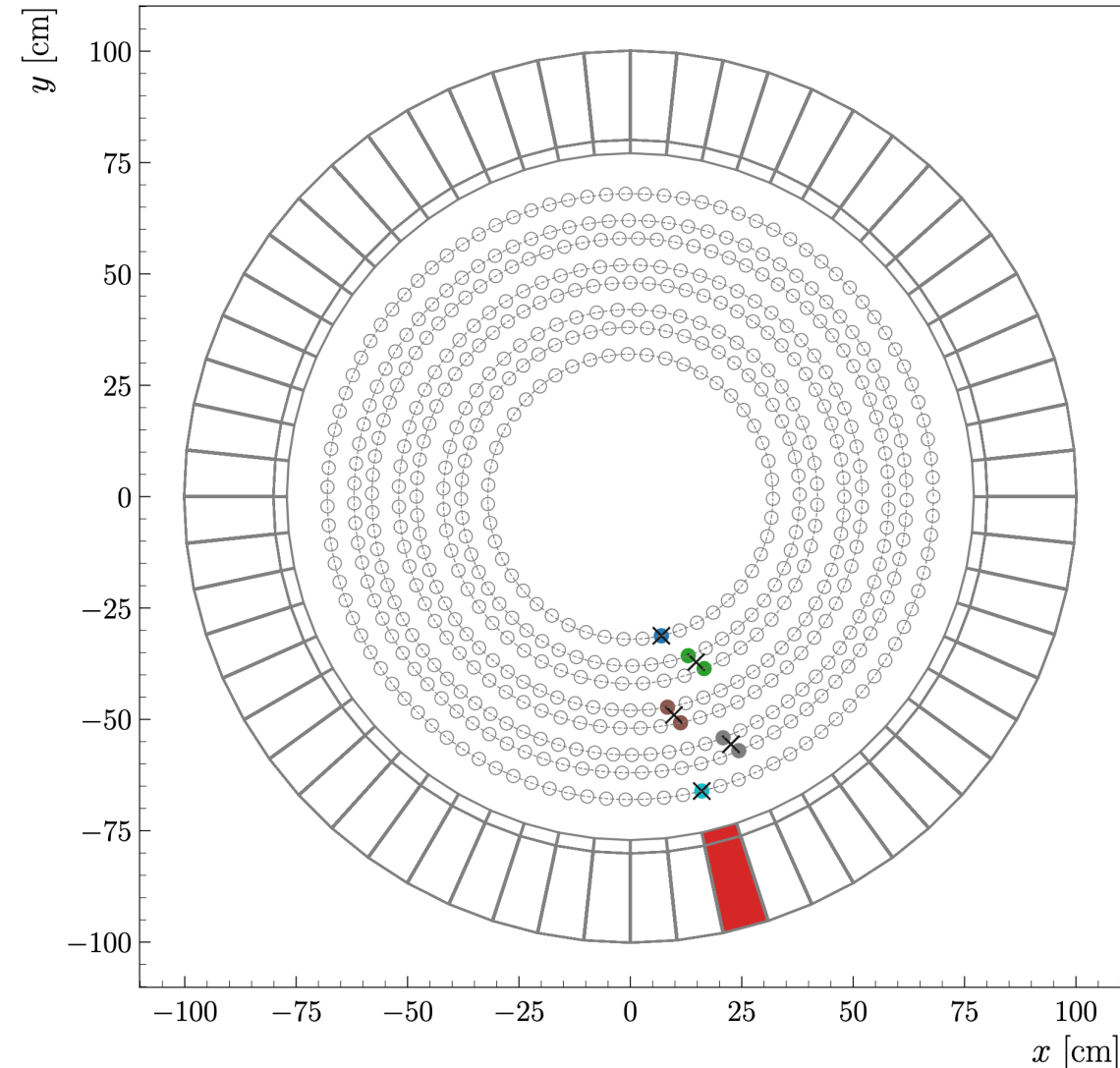
- Extract p_T broadening
- Extract $\cos(\phi_h)$ and $\sin(\phi_h)$ modulation
- Retrain the AI for photon identification for RG-D



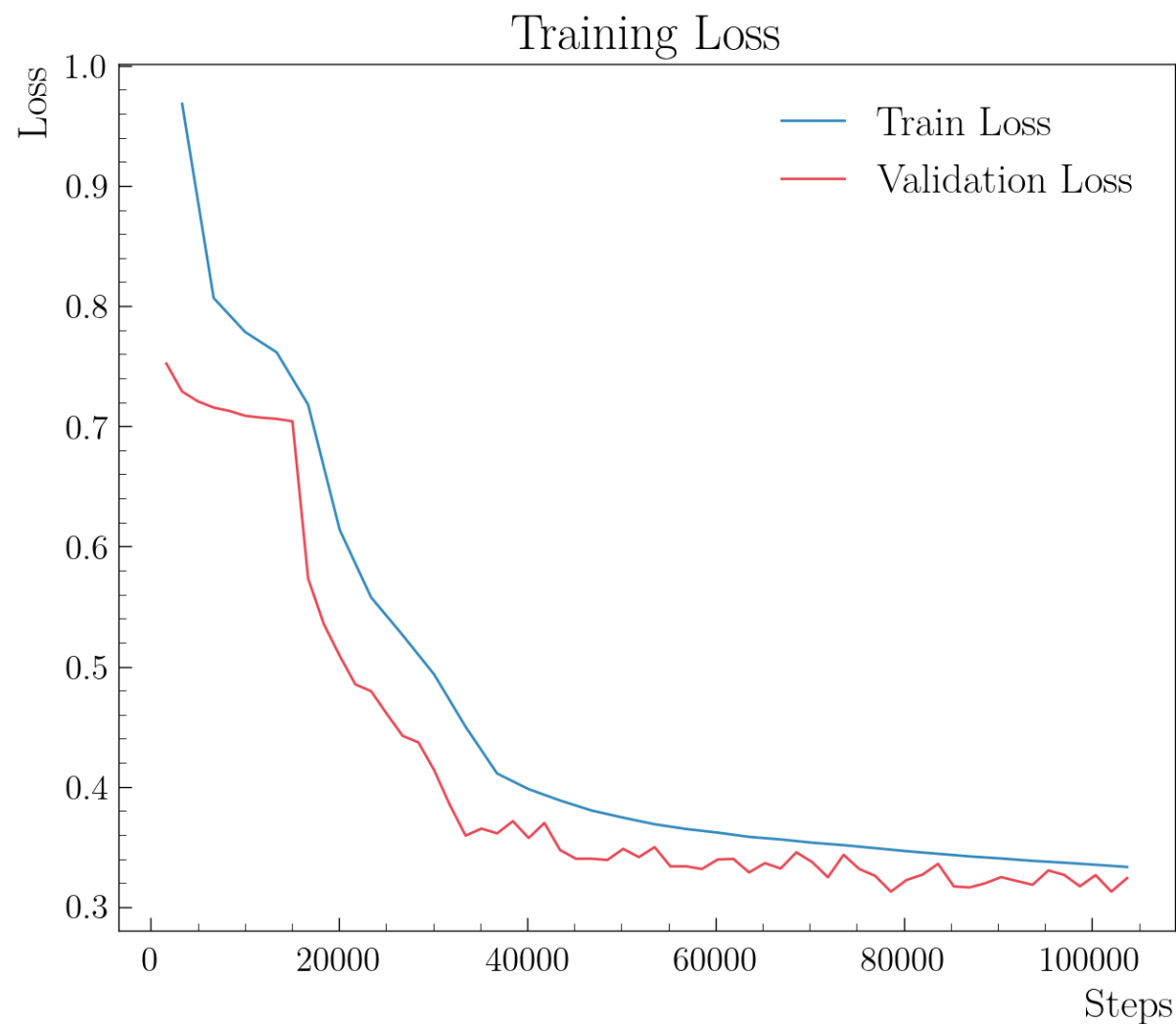
- First, focus on the track matching:
 - Identifying which bar and edge should be hit based on the AHDC information (hit base only)
- Then, work on the momentum prediction with a different network
- Use an MLP to do the prediction
- Right now, still under development. I am trying to find the best architecture for the model.

AI for track matching

- Goal: with an AI, predict which bar and edge should be hit based on the clusters (or inter-clusters):
- Inputs: 4 of 3-D points (cluster: x, y, z) (in total 12 numbers)
- Output: sector, layer, wedge
- For the training, 5M events, and select them with a good track ($\text{pred} > 0.2$), and only one bar and wedge hit

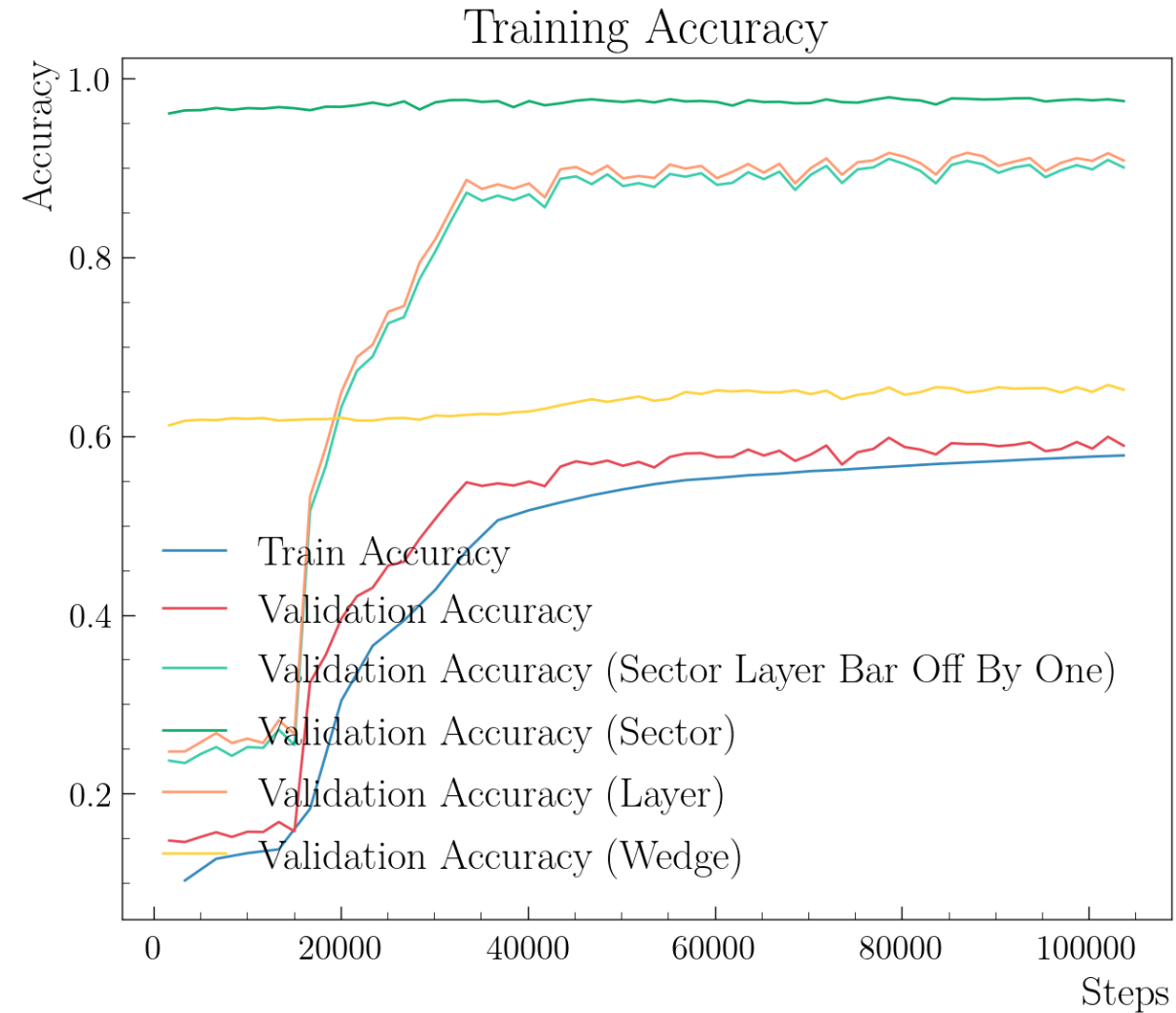


- Plot the loss function:
 - Error between the prediction and the truth.
 - $\frac{1}{3} * \text{sector_error} + \frac{1}{3} * \text{layer_error} + \frac{1}{3} * \text{wedge_error}$
 - No overfitting: the train and the validation loss stay stable.



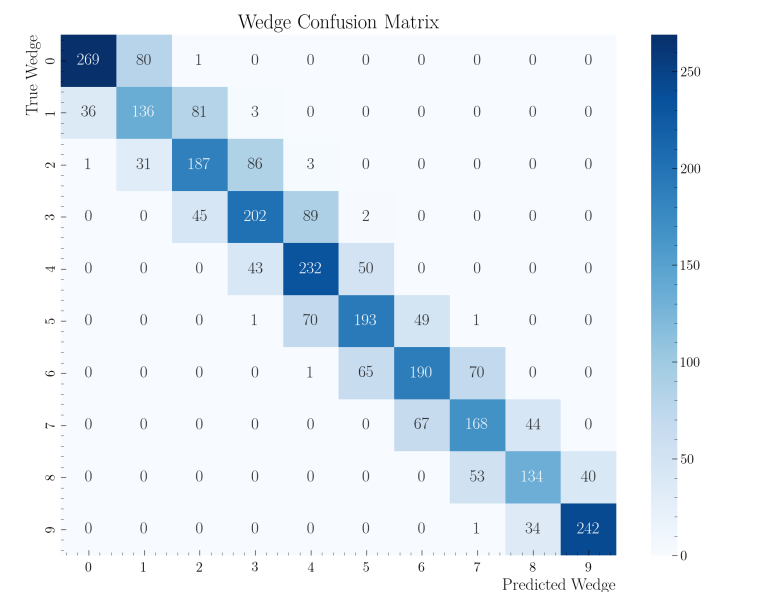
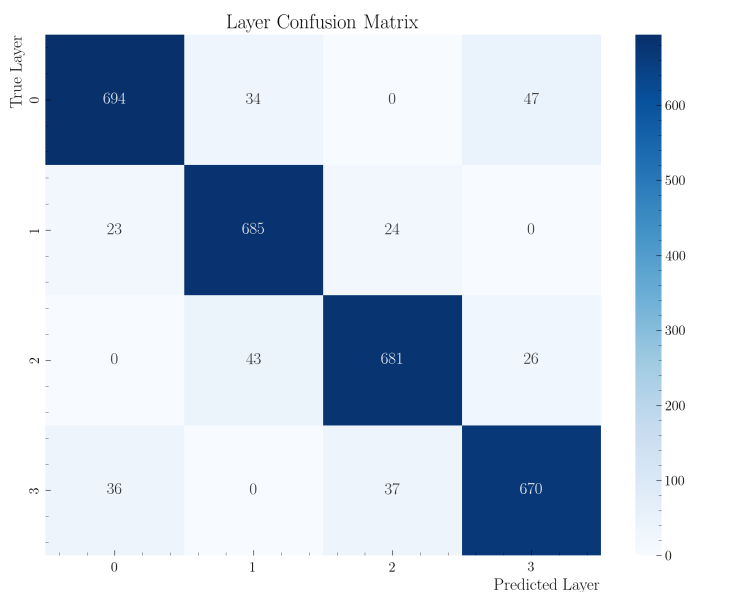
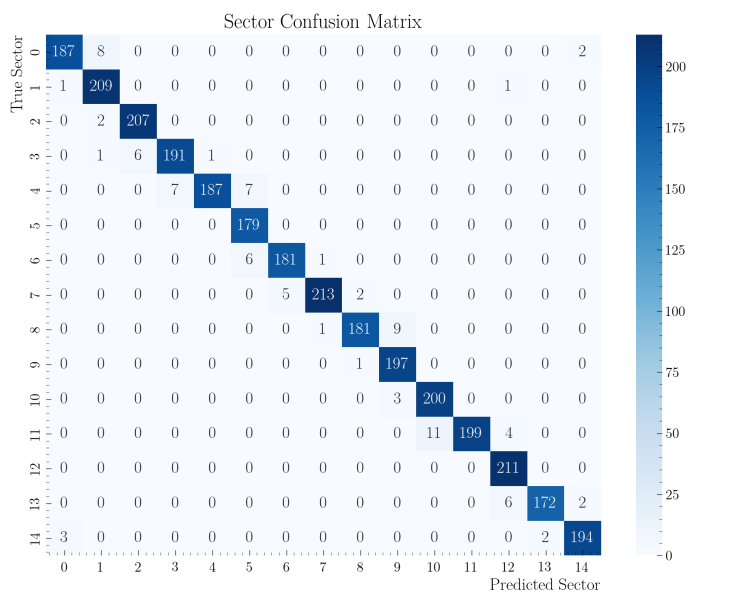
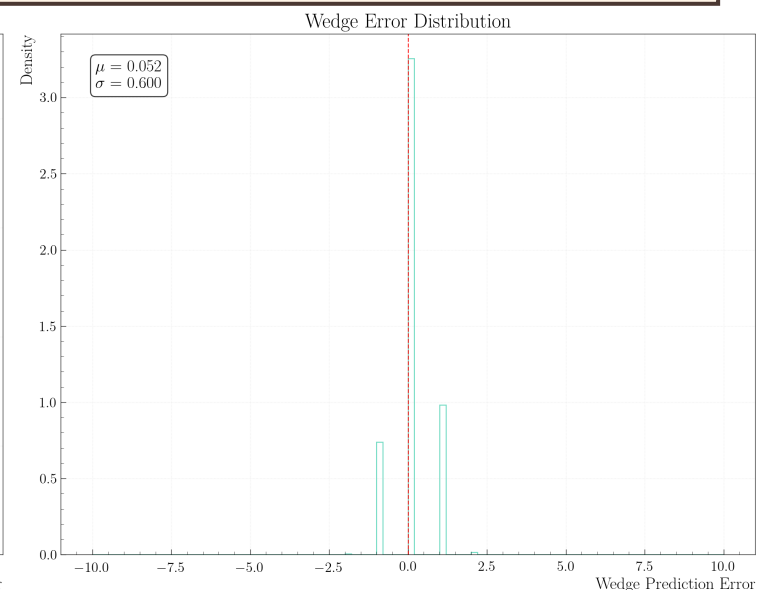
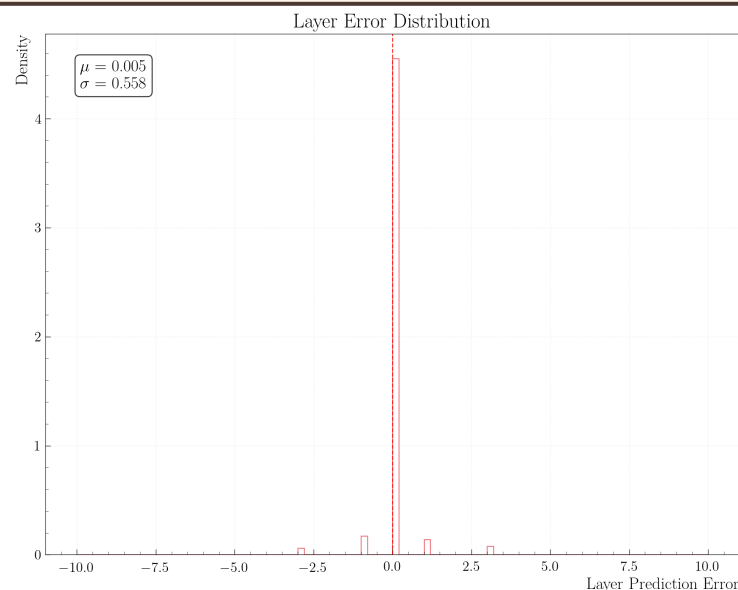
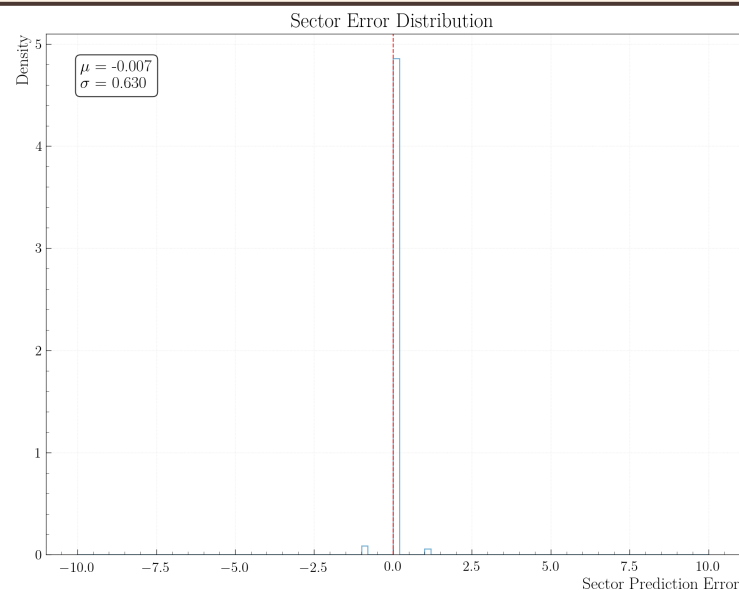
AI for track matching

- Plot the accuracy (how often the model predicts the true value):
 - If the model predicts the true value every time, the accuracy is 1
 - If the model never predicts the true value, the accuracy is 0
- Train/Validation accuracy is when the model predicts the correct sector, layer, and wedge.
- The Sector Layer bar of by one is the accuracy when the model predicts the correct sector, layer but the wedge is the correct one or off by one.



AI for track matching

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- swif-gui:

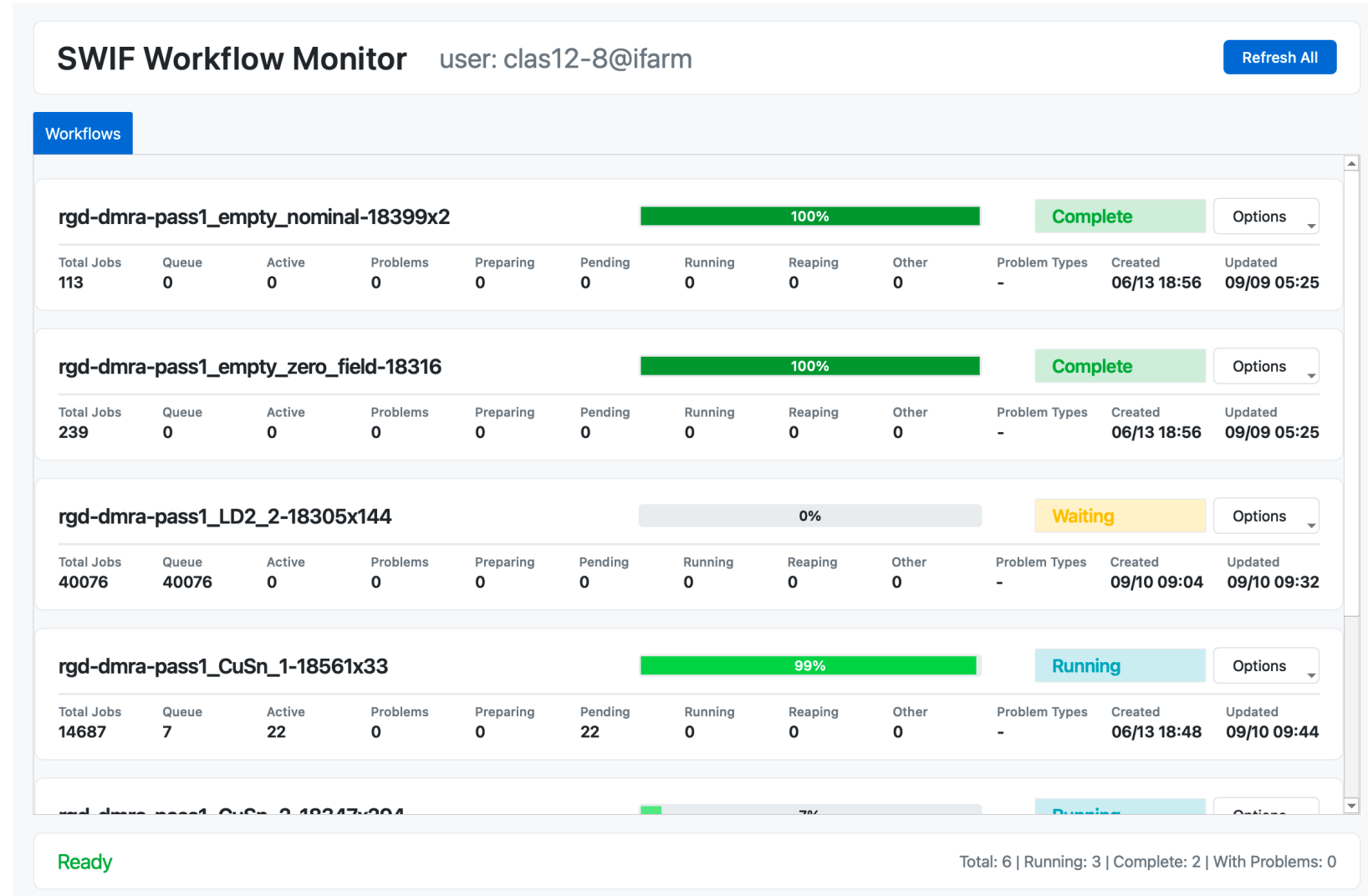
Summary

Name	Complete	Jobs	Undispatched	Dispatched	Preparing	Pending	Running	Reaping	Other	Problems	Types	Create	Update	
rgd_simulation_pion_test	0.00%	20	10	10	0	0	0	0	0	10	SITE_LAUNCH_FAIL	09/11 17:01:08	09/13 18:01:51	Options

Workflow monitor

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- swif-qt:
- Rewrite the GUI with Qt during the weekend (last 3 weekend)



Workflow monitor

- swif-qt:
- Rewrite the GUI with Qt during the weekend (last 3 weekend)

SWIF Workflow Monitor user: ouillon@ifarm [Refresh All](#)

Workflows

rgd_simulation_pion_test

0%

Has Problems

Options

Total Jobs	Queue	Active	Problems	Preparing	Pending	Running	Reaping	Other	Problem Types	Created	Updated
20	10	10	10	0	0	0	0	0	SITE_LAUNCH_FAIL	09/11 17:01	09/13 18:01

Ready

Total: 1 | Running: 1 | Complete: 0 | With Problems: 1